

Final Design: W.I.L.L-E
(Warman Industrial Load Lifting Experiment)
Weir-Warman Competition
Team 77

James Robinson	

Executive Summary

The outcome of the project has been a successful one, through delegation, consideration and testing. A successful project robot was able to be constructed. Over the course of the 6 weeks of testing designing and producing WILL:E, we redesigned and overcame several obstacles relating to all aspects of the robots functionality and accessibility. Conclusively our robot consistently demonstrated high performance in both design and functionality, excelling in tasks such as maneuverability, ball manipulation, and system modularity. The success reflected our strong teamwork, innovative problem-solving, and thorough testing and construction process



Figure 3: WILL:E testing new ball collection intake system

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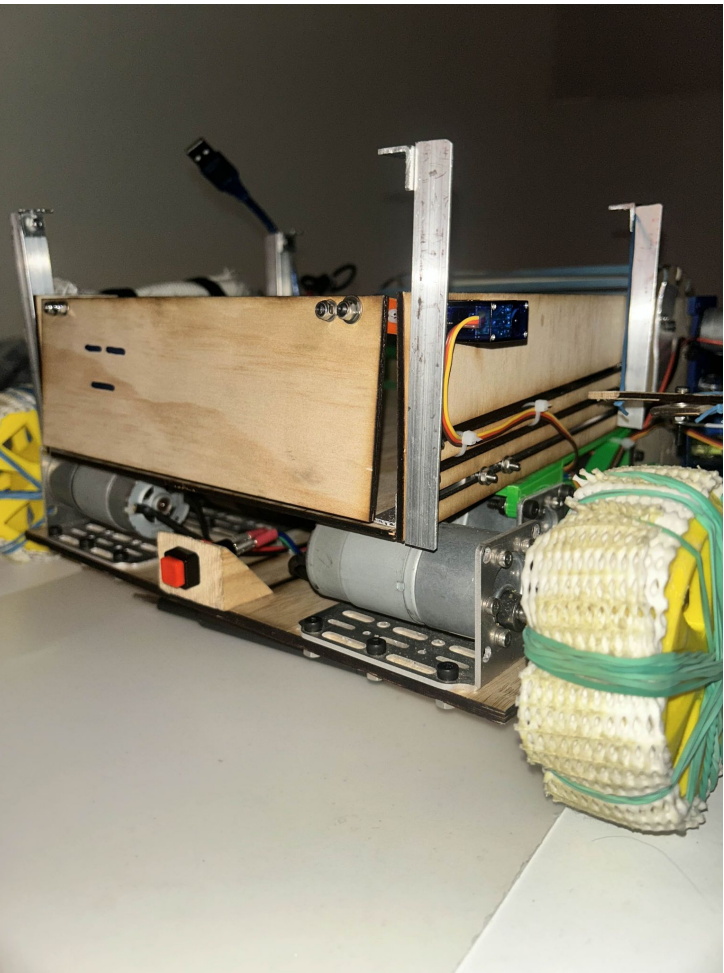
1.0 Introduction and Executive Summary

Objective Outcome

Overall, the team performed up to the necessary expectations and achieved a high enough score through the prerecorded video of the robot accomplishing all tasks within the necessary time frame. A higher more proper score was not able to be achieved on the day of the competition due to damages and technical difficulties to do with the collection system hindering a proper second run.

Competition Details

The Weir Warman Competition for 2025 involved students and teams to design and construct a ball gathering robot able to pick up and contain 3 balls of differing size and distance. Having collected the 3 necessary balls, the teams were challenged to have the robot to traverse a see-saw. After passing the see-saw the robot needs to deposit the contained balls within a final collection zone before maneuvering into a final end zone for the robot.



Our Team

Our team is composed of 5 passionate engineering students driven to engage and gain hands-on experience and advancing our skills in mechatronics, electrical, software and mechanical engineering. With diverse backgrounds in mechanical engineering, electronics, and computer science experience, we collaborate to solve real-world challenges through creativity, technology and humour.

Figure 1: WILL:E Final Rear Emplacement Kinetic (FREK)

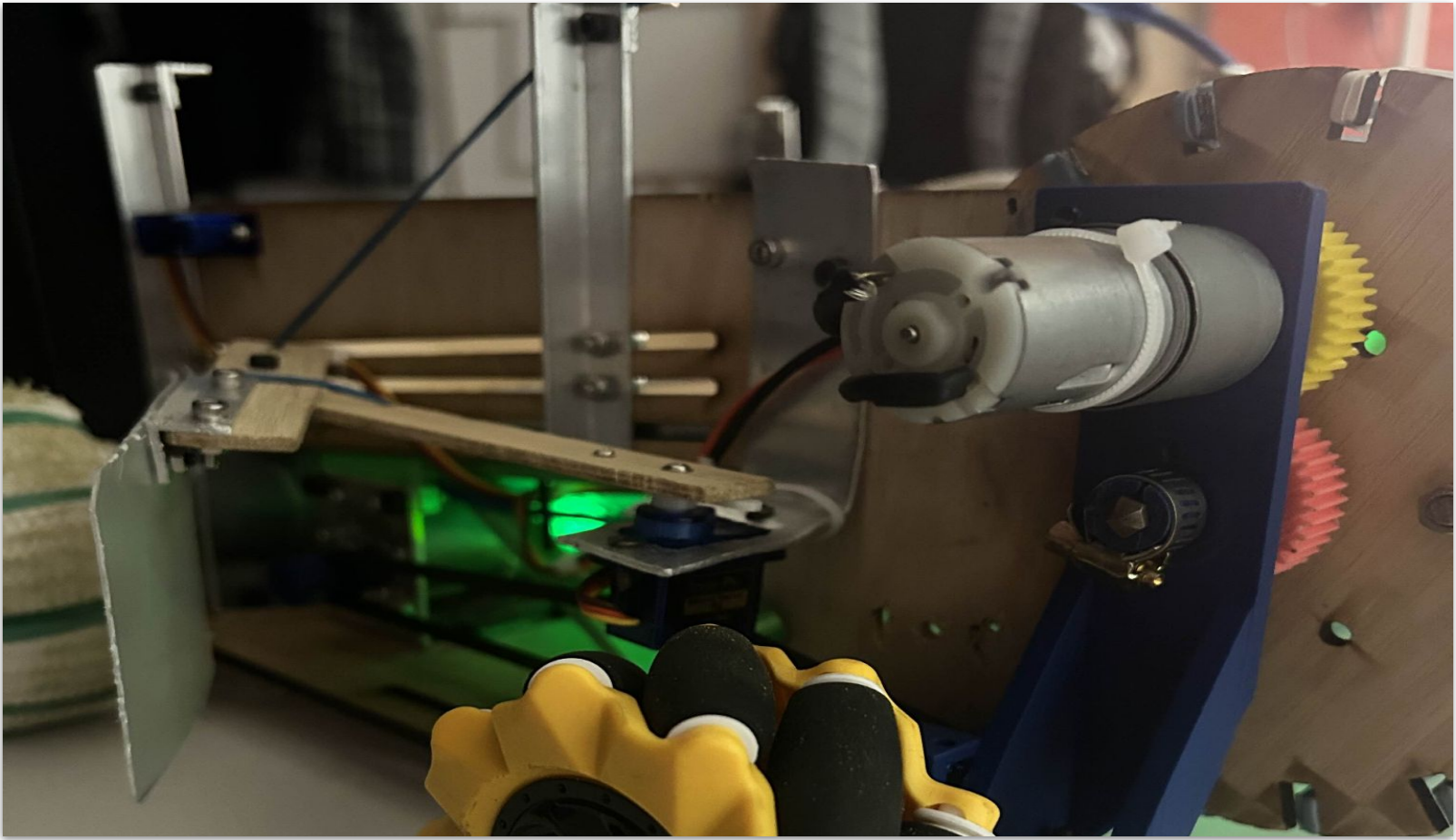


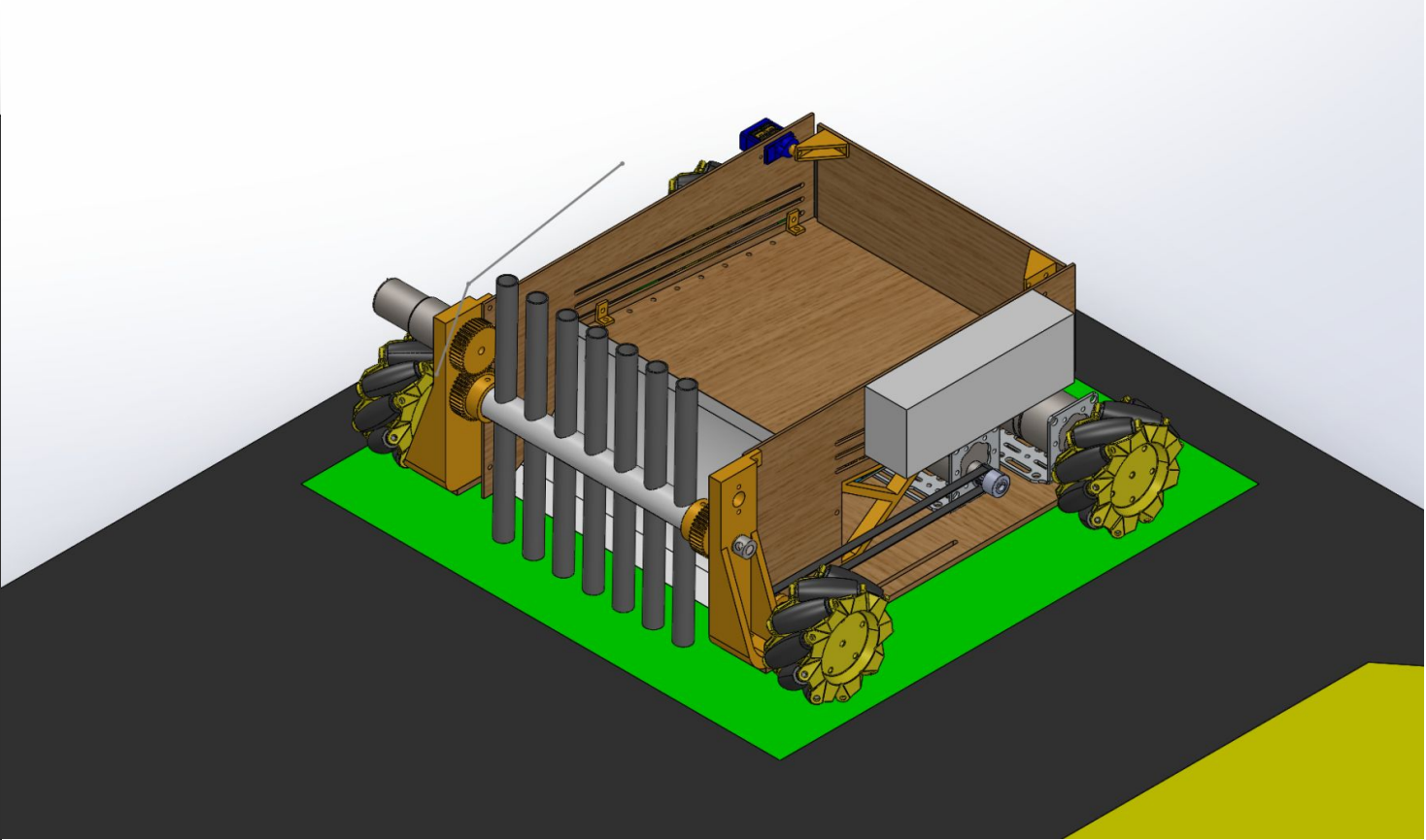
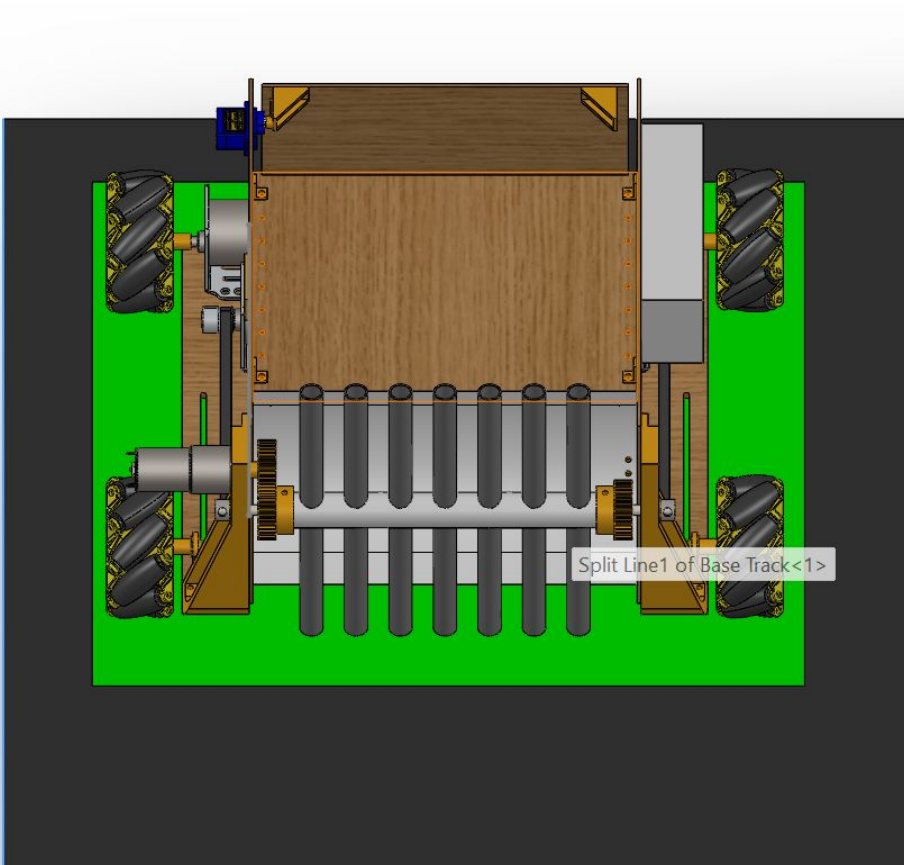
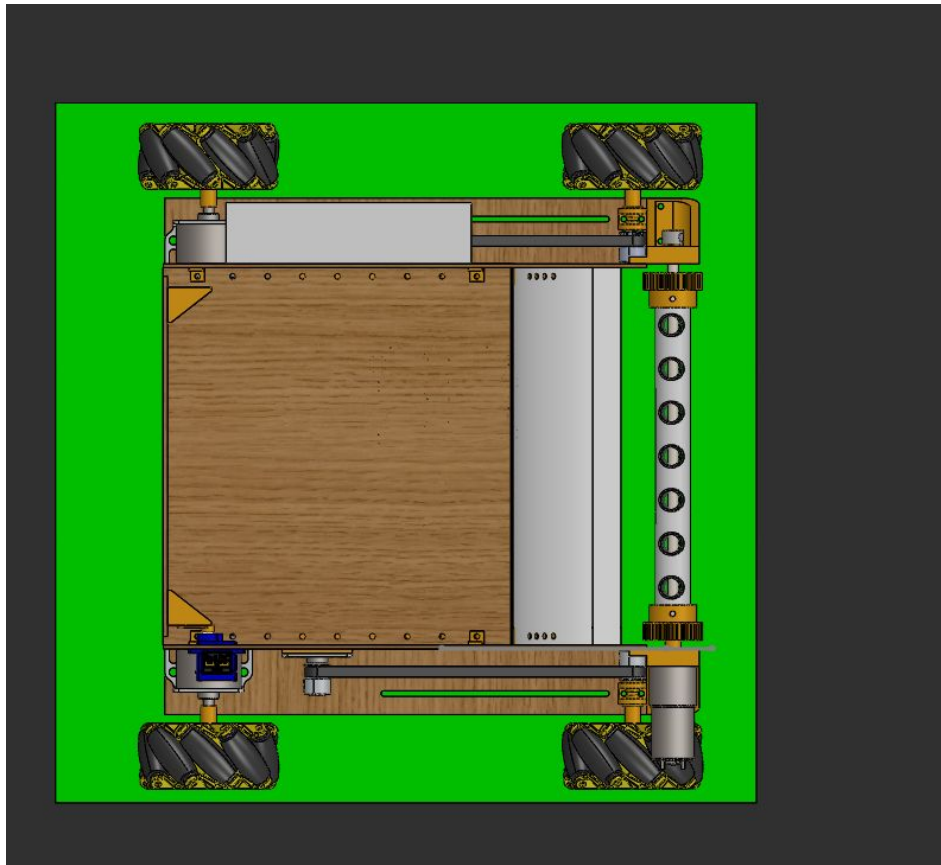
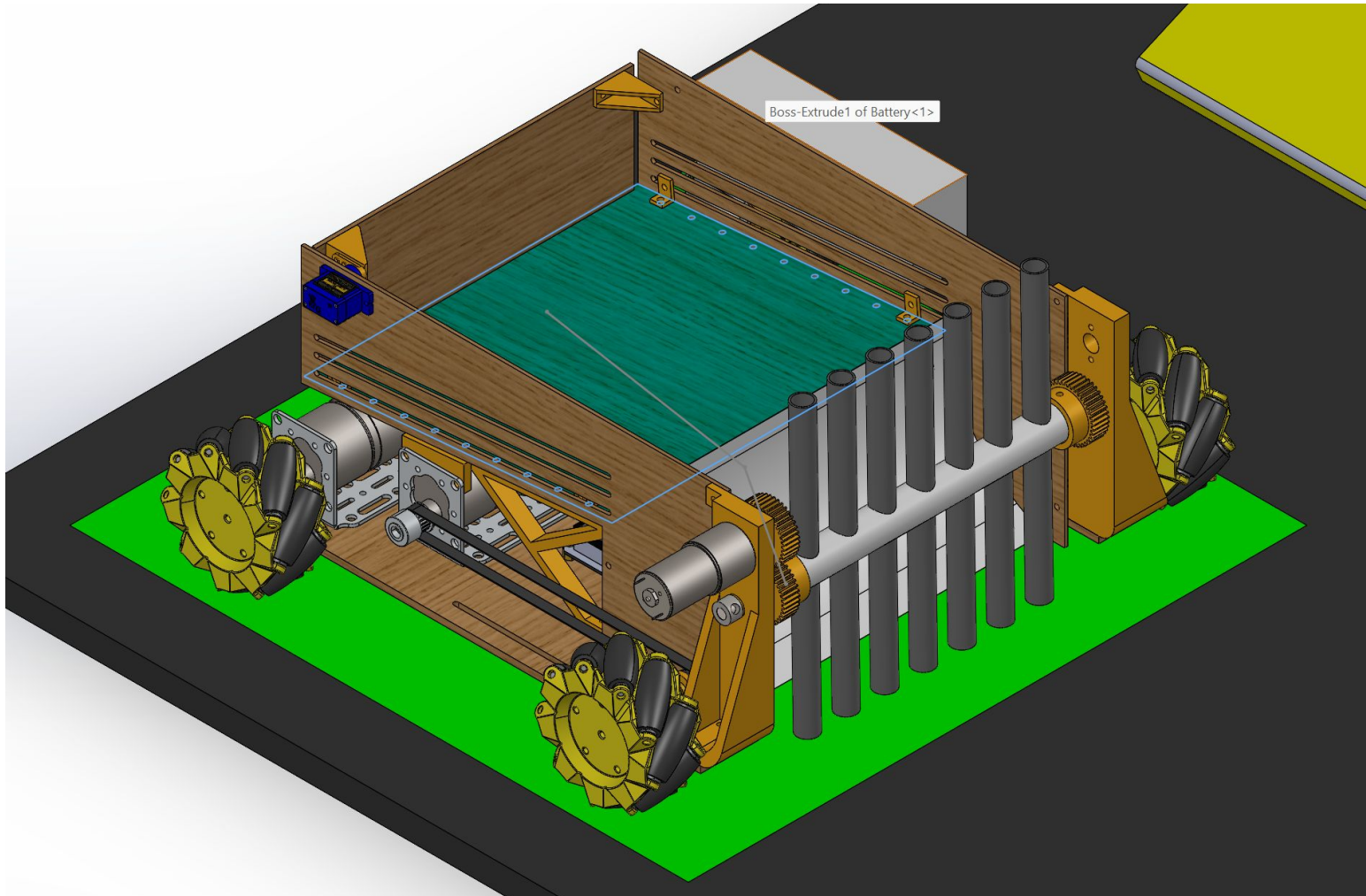
Figure 2: WILL:E side flapper profile

Contents

The following slides performs and displays various stages of the robots creation. Slides 5 - 13 will depict the robot in its various stages throughout the designated track., comparing the preliminary prototype to the final assembly in the same stages of the same track. Slides 14 - 18 will illustrate the various design changes that went through with the robot, from wheel and grip considerations to the intake system and will illustrate the reasons and methods undertaken to redesign each system to perform to expectation. Slide 19 discusses the project reflection and the lessons learned from the entire project while Slide 20 discusses the financial commitments made to ensure the robot had the best possible opportunity to succeed within the competition.

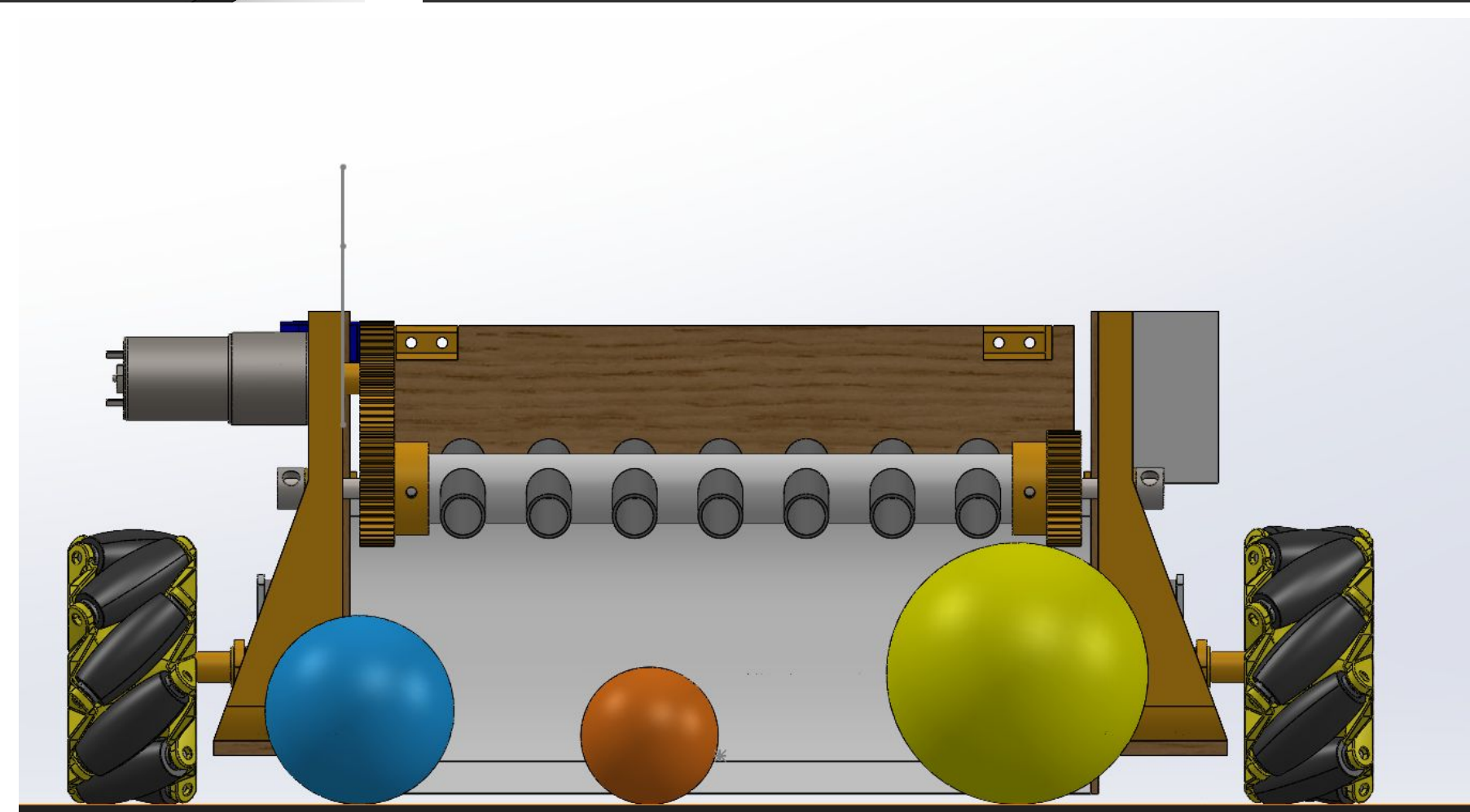
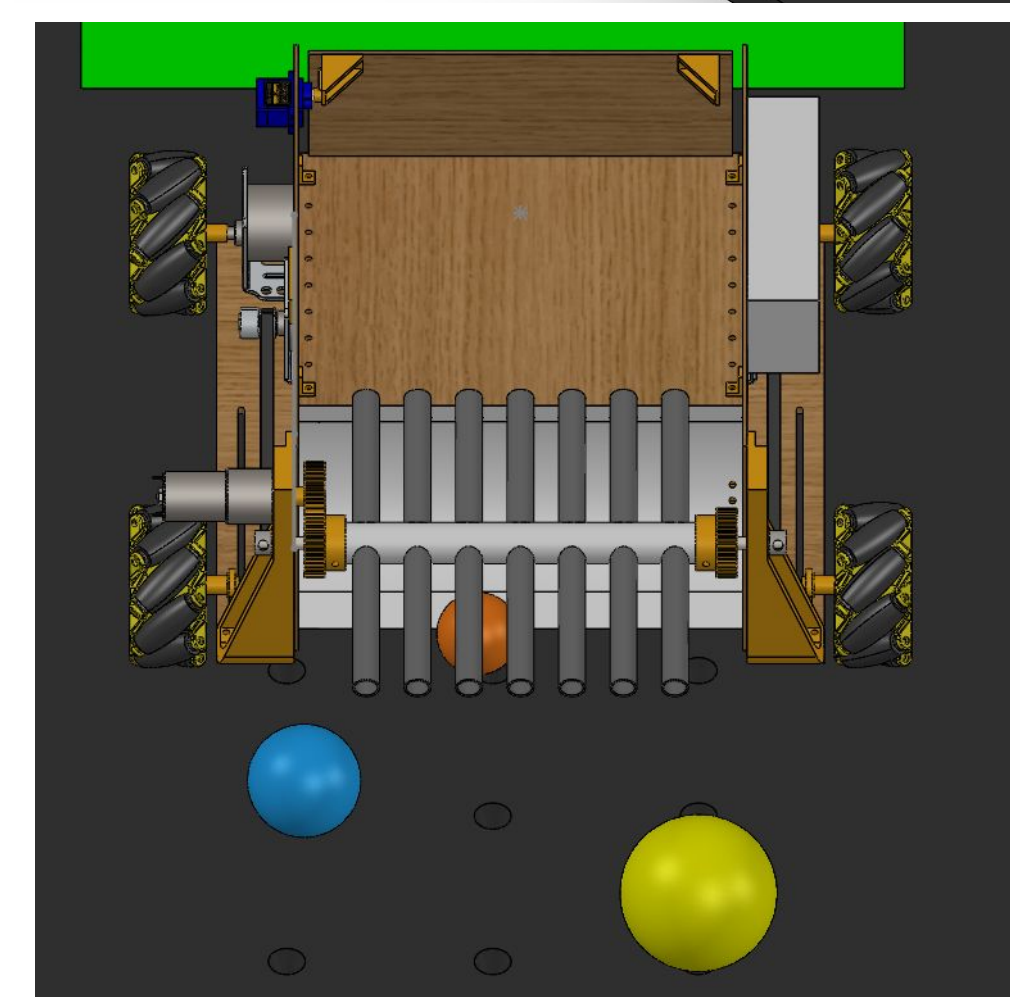
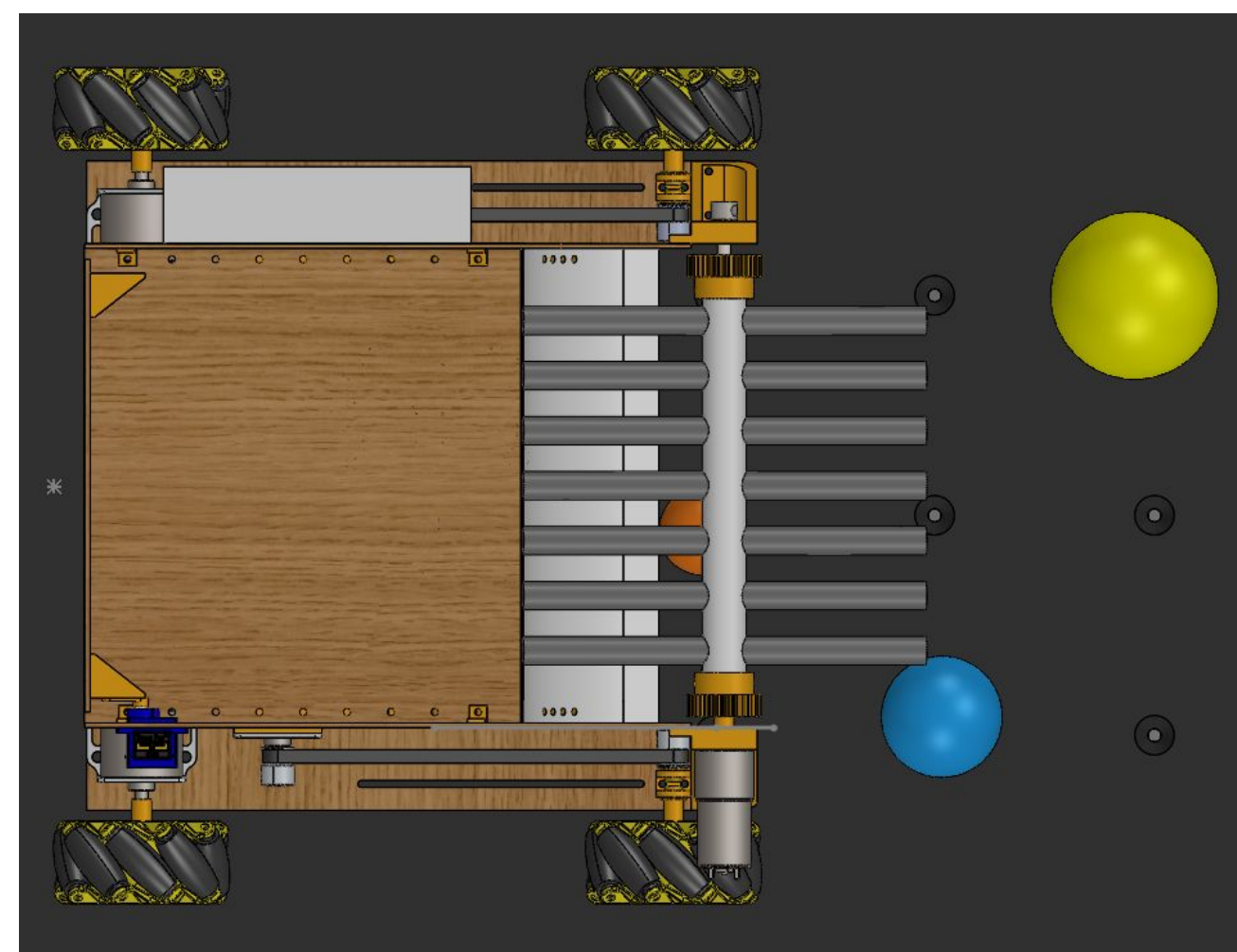
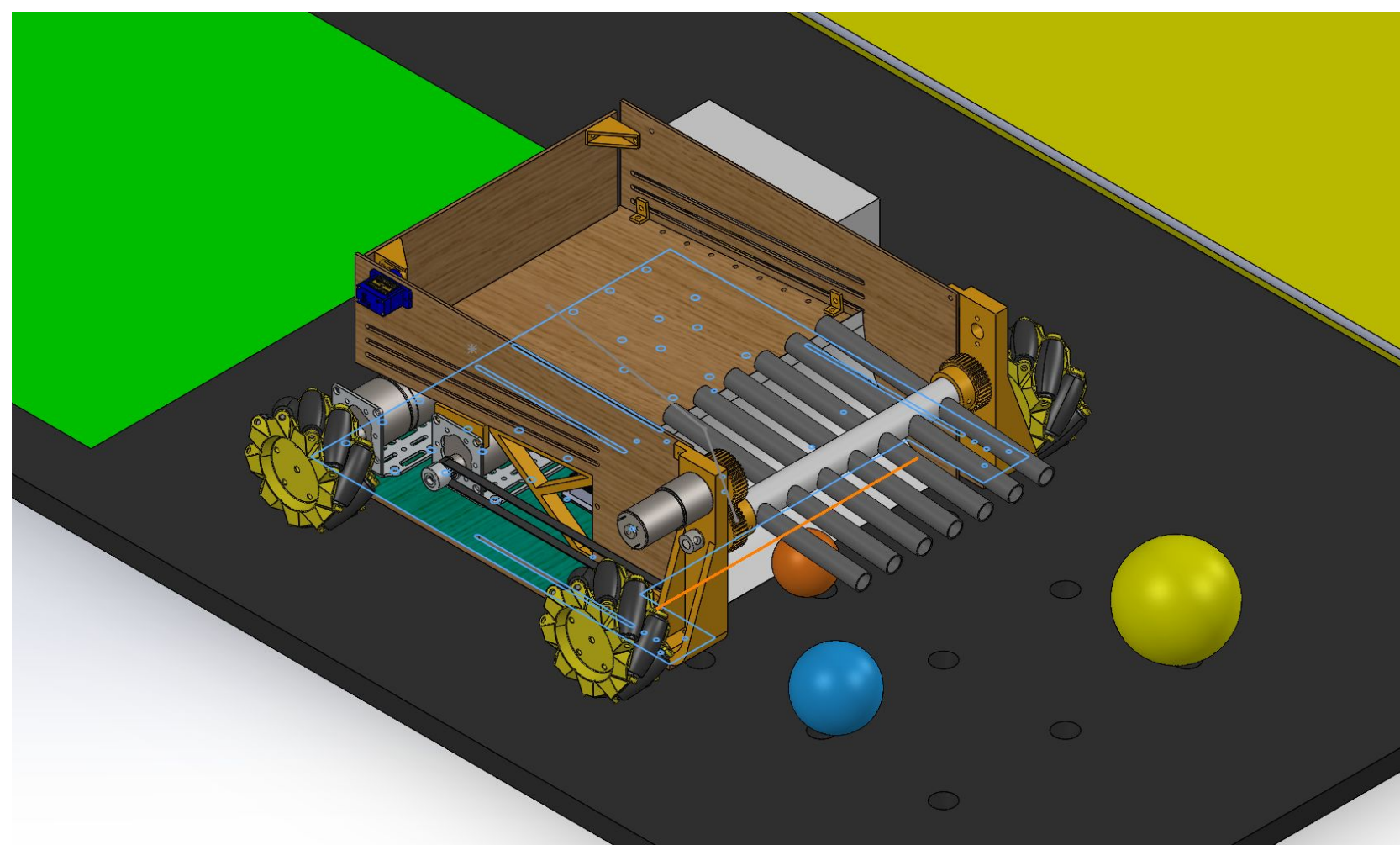
2.1 CAD Images (System Set-up and ready for start)

PHASE 1



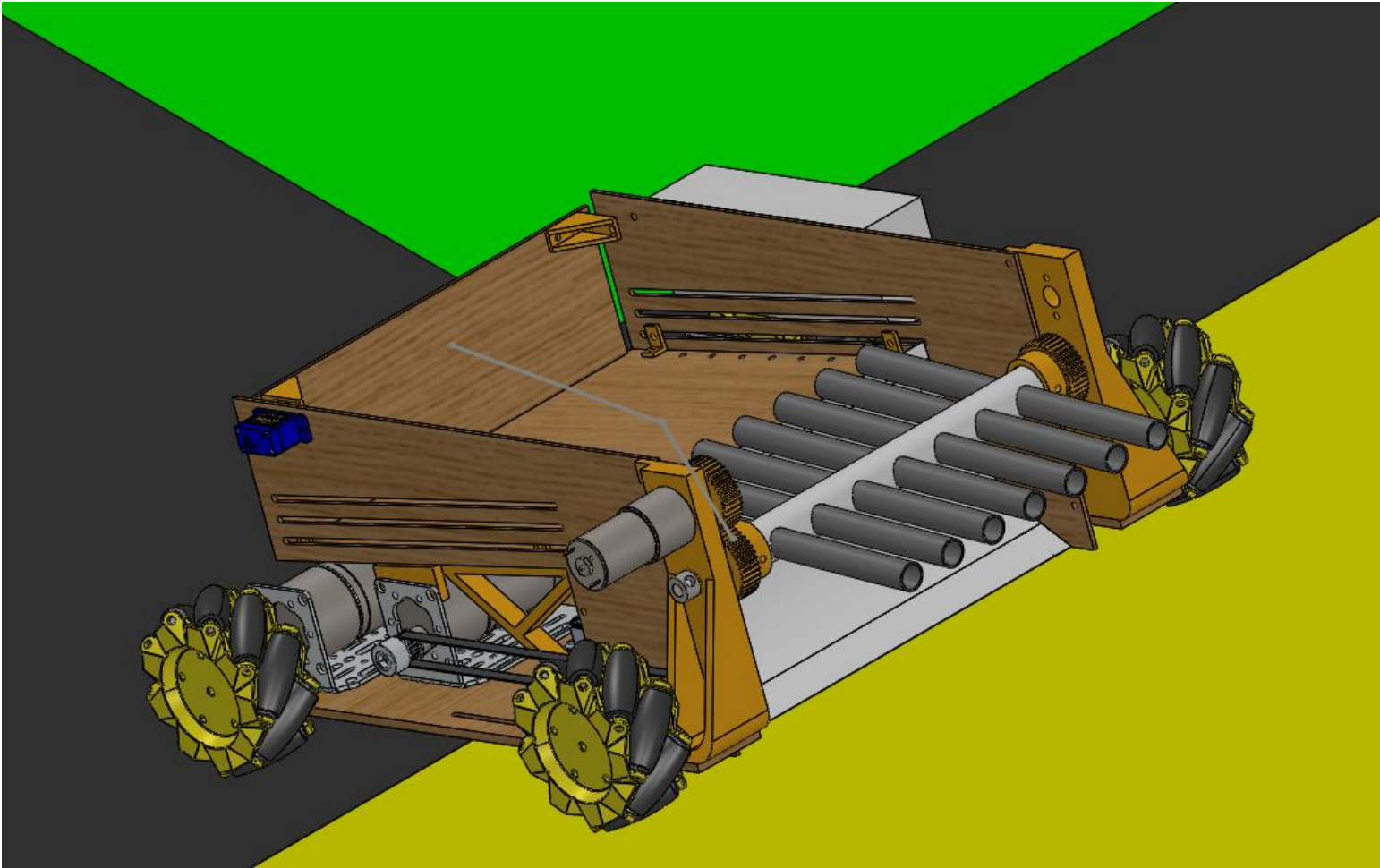
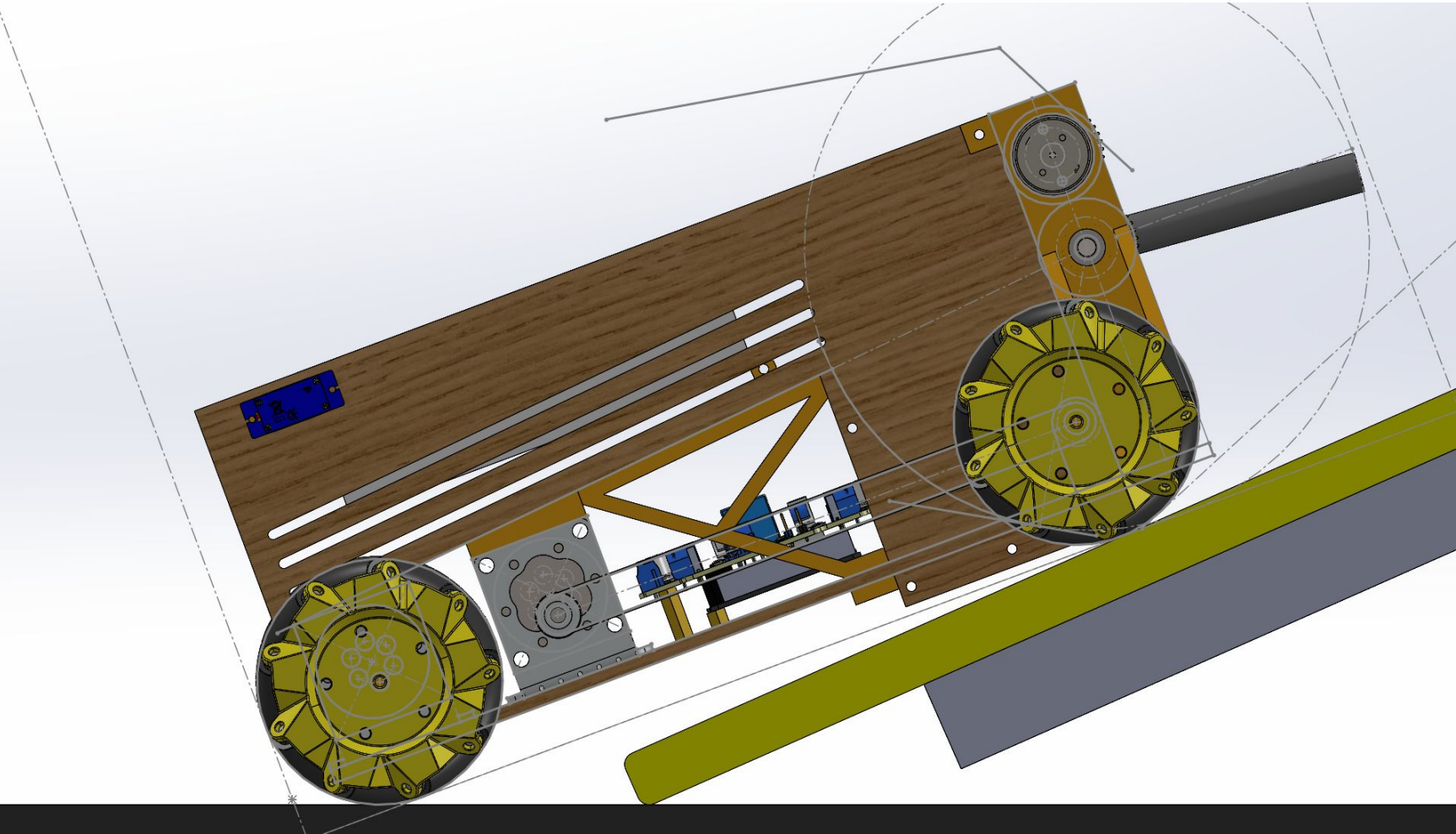
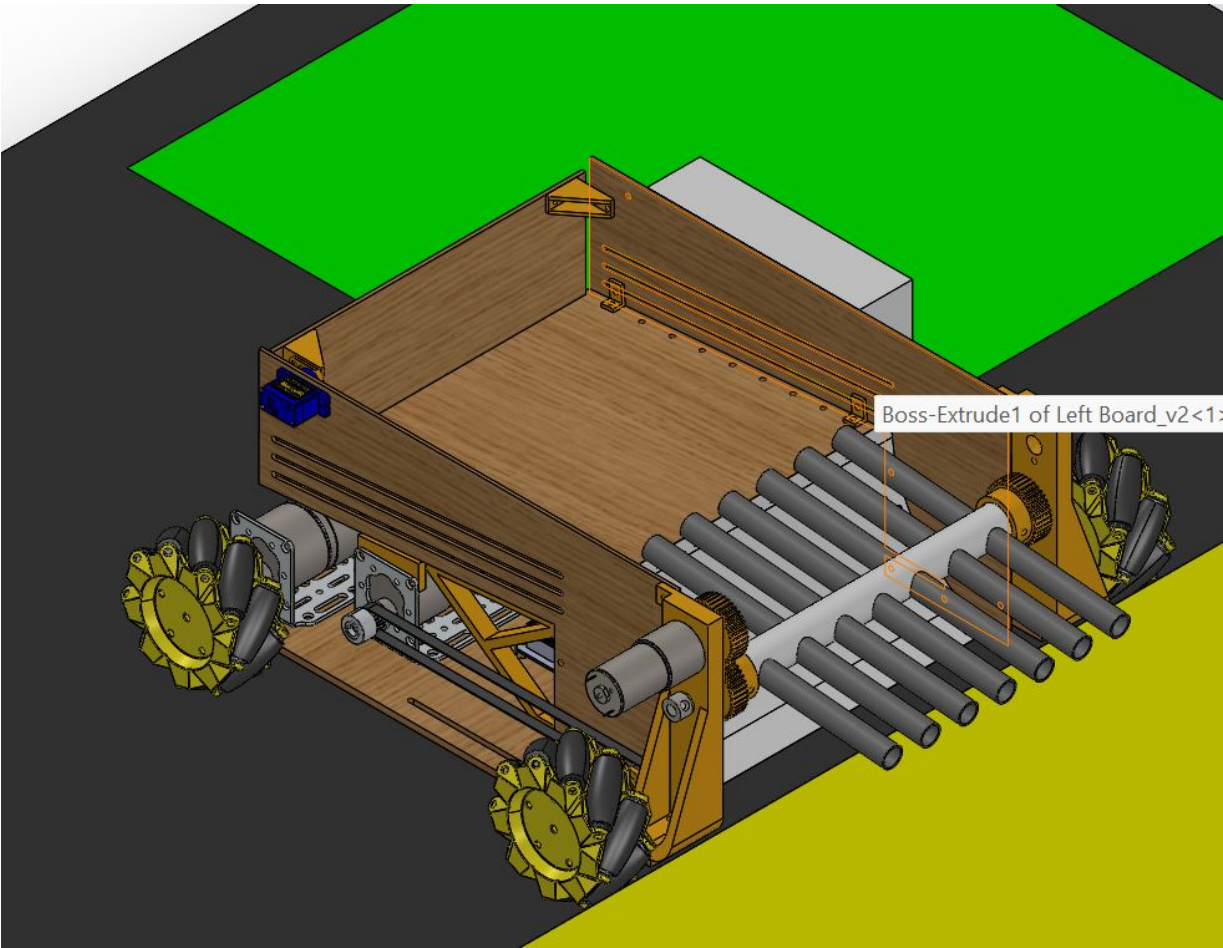
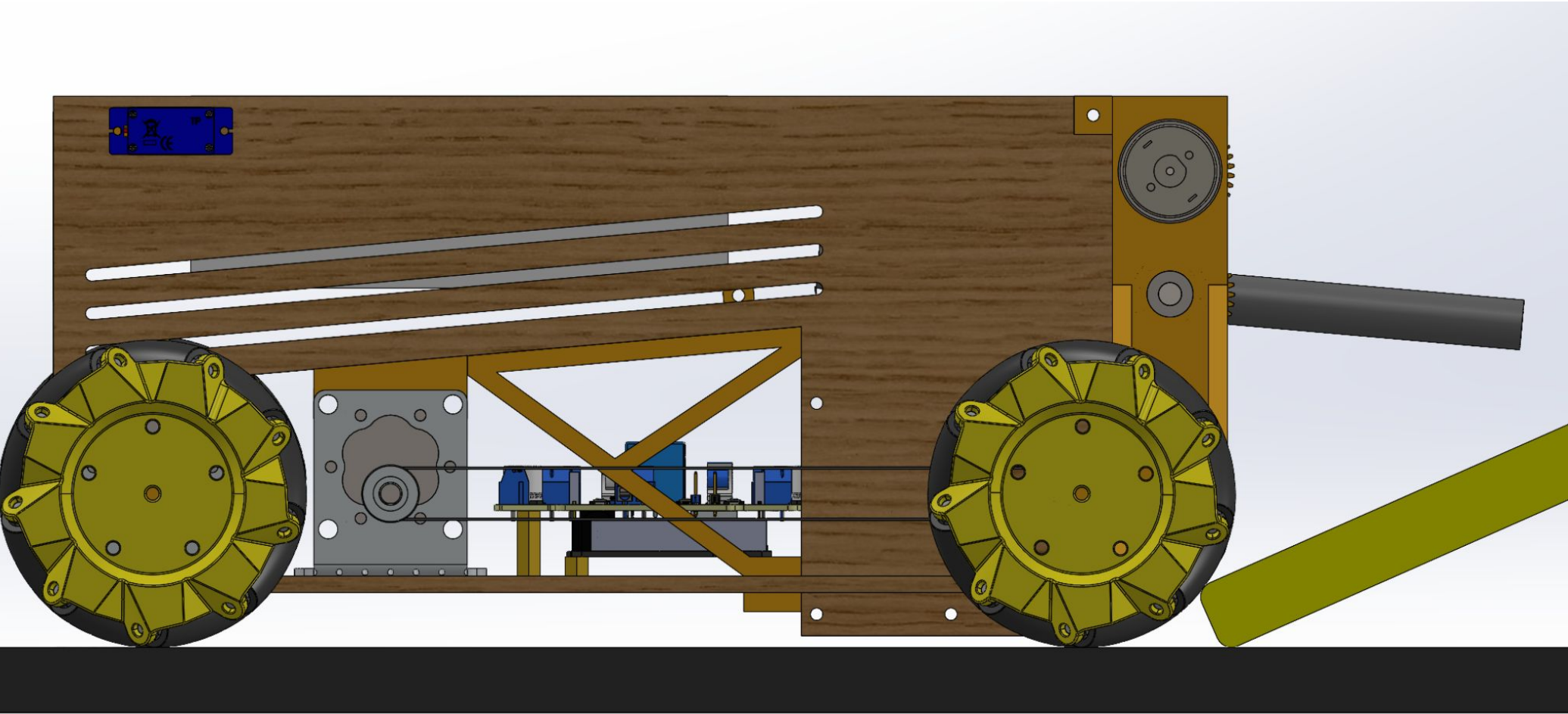
CAD Images (Pick-Up Ball)

PHASE 2



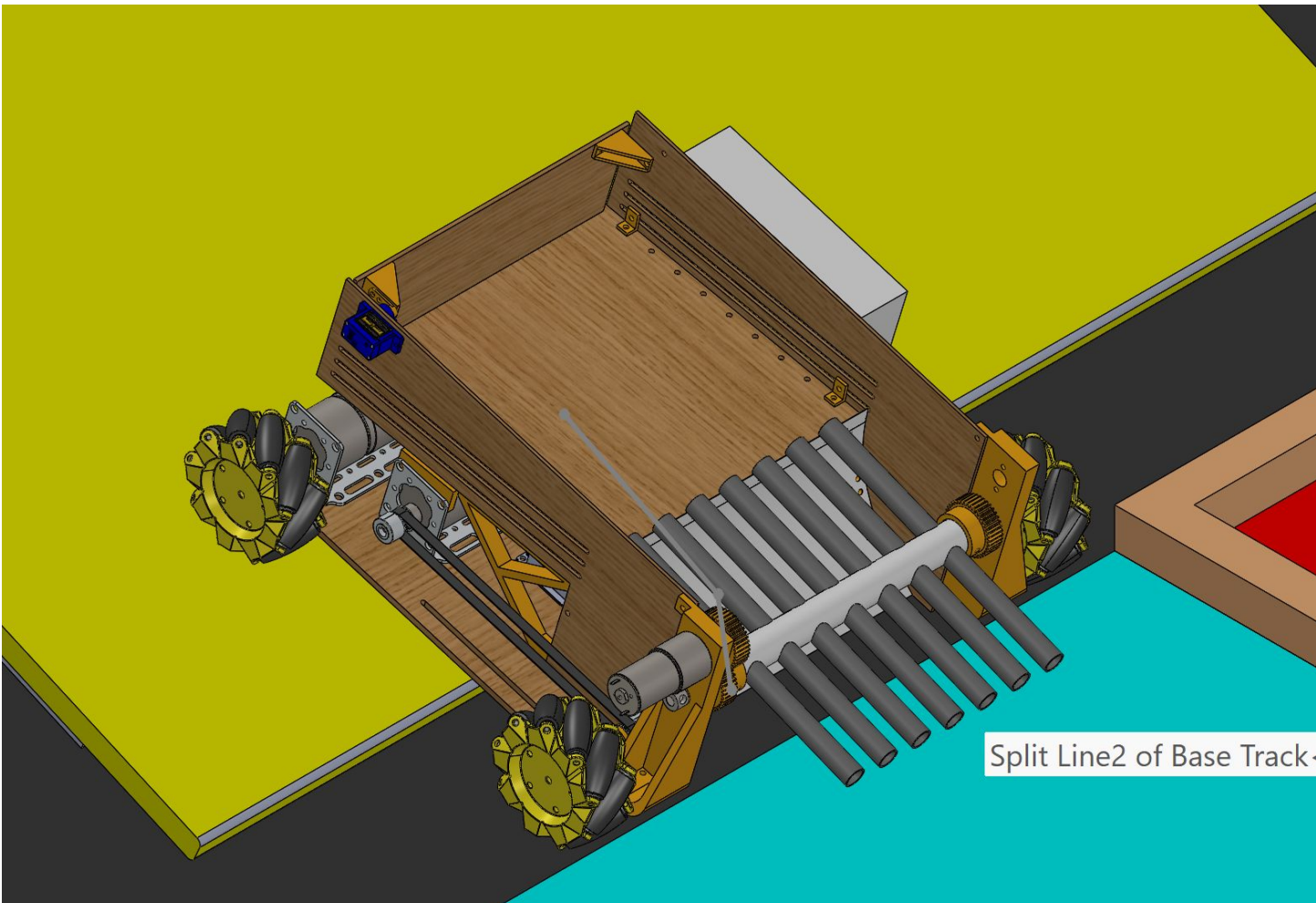
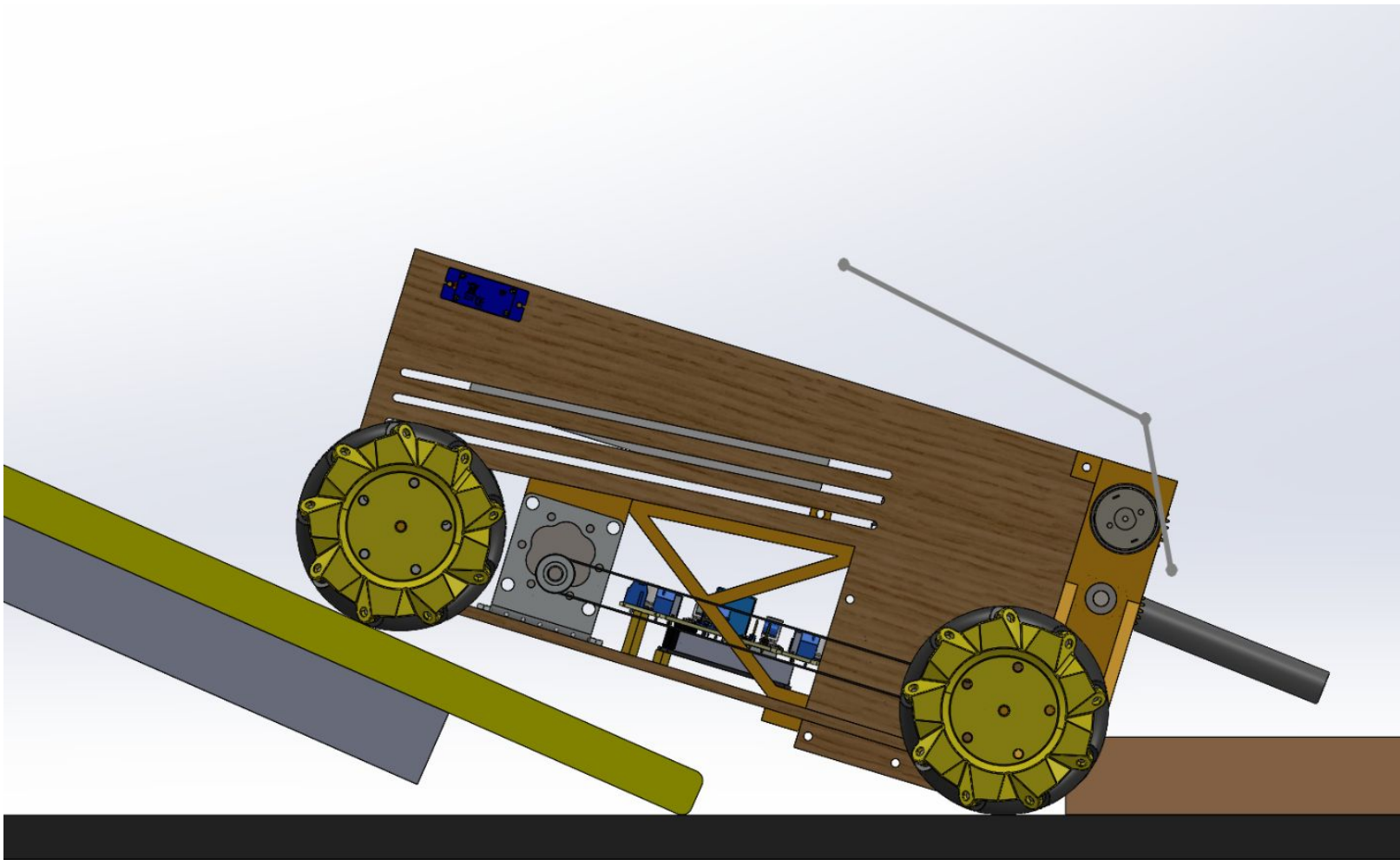
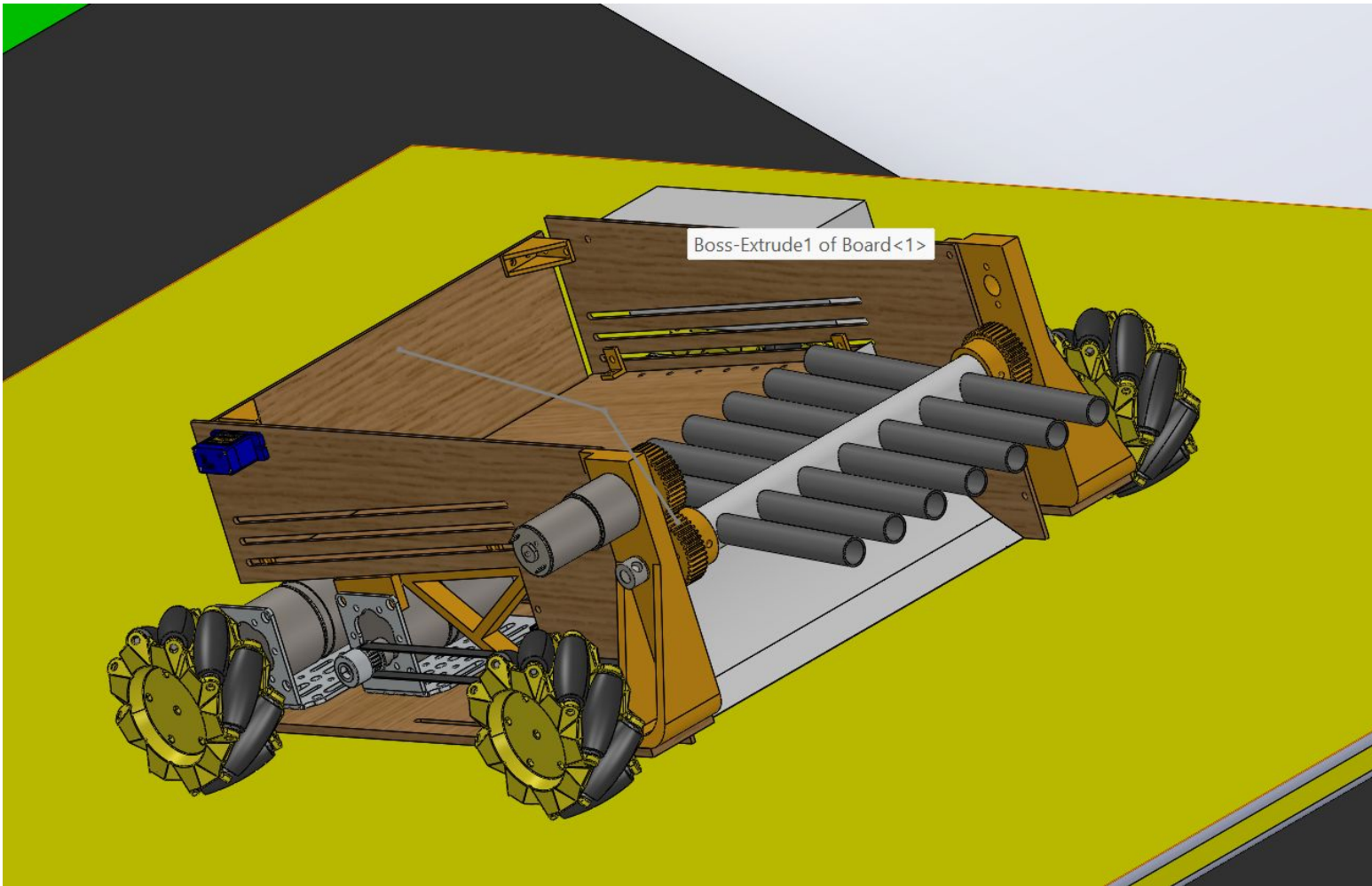
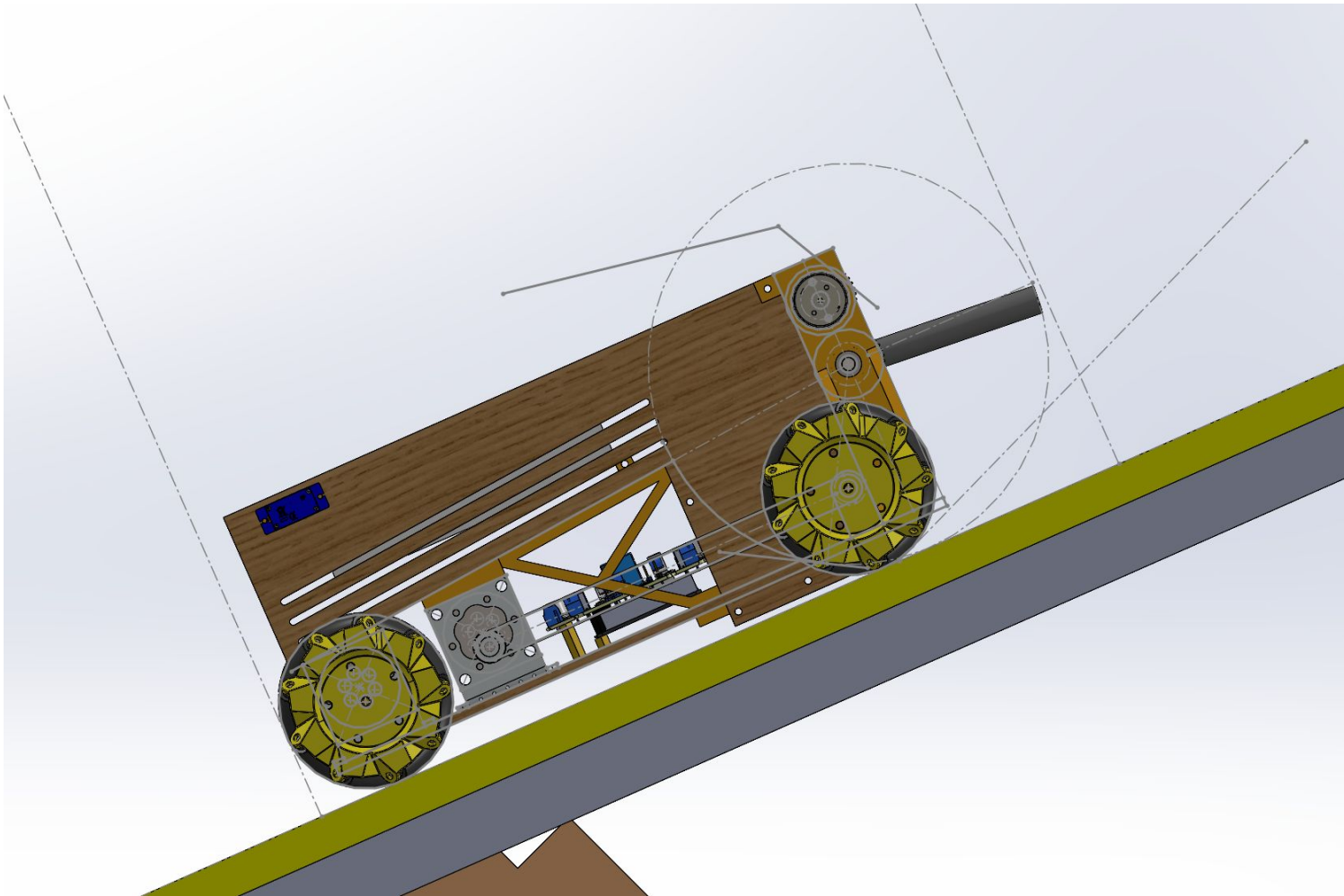
CAD Images (Traverse Seesaw)

PHASE 3



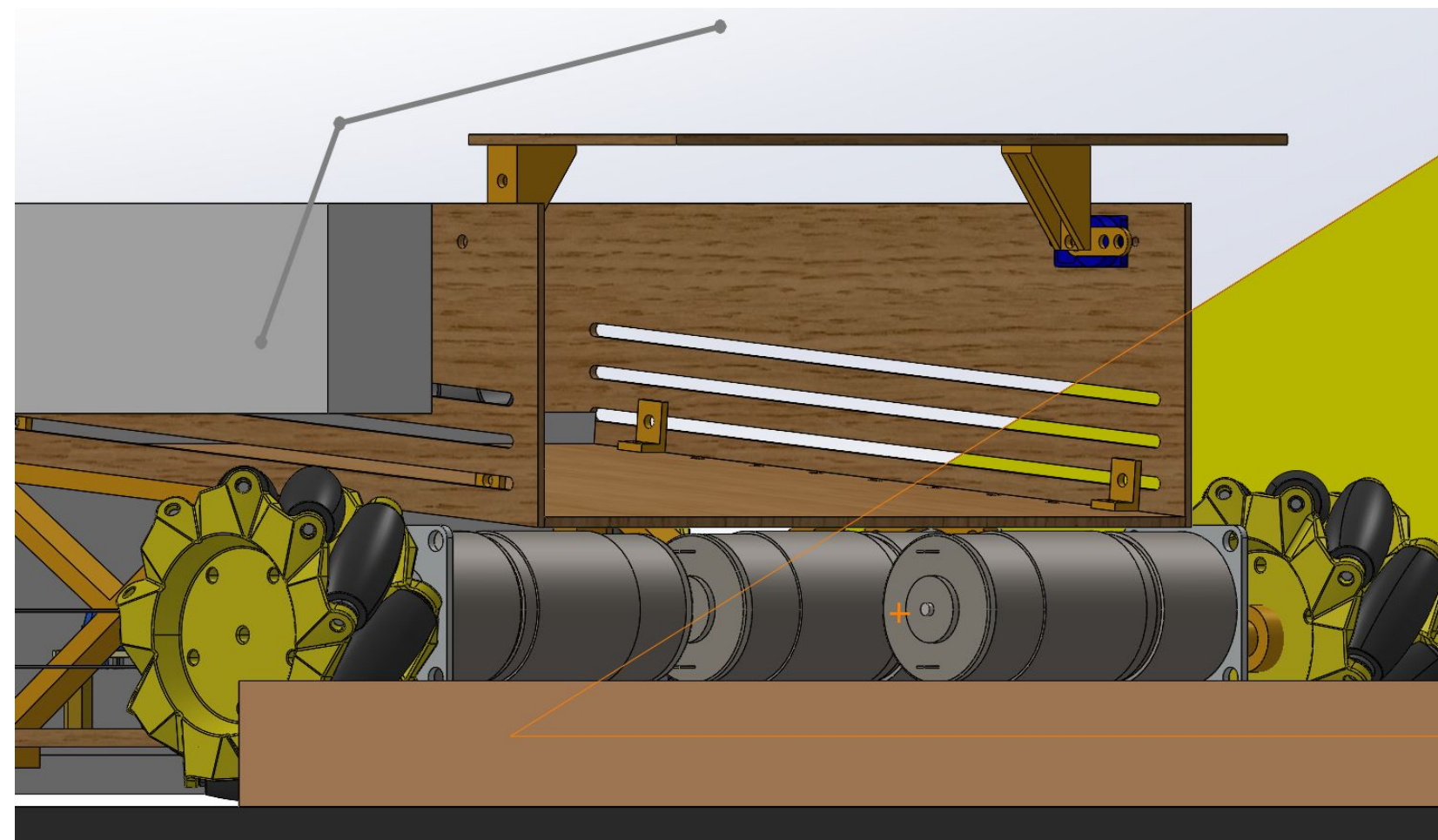
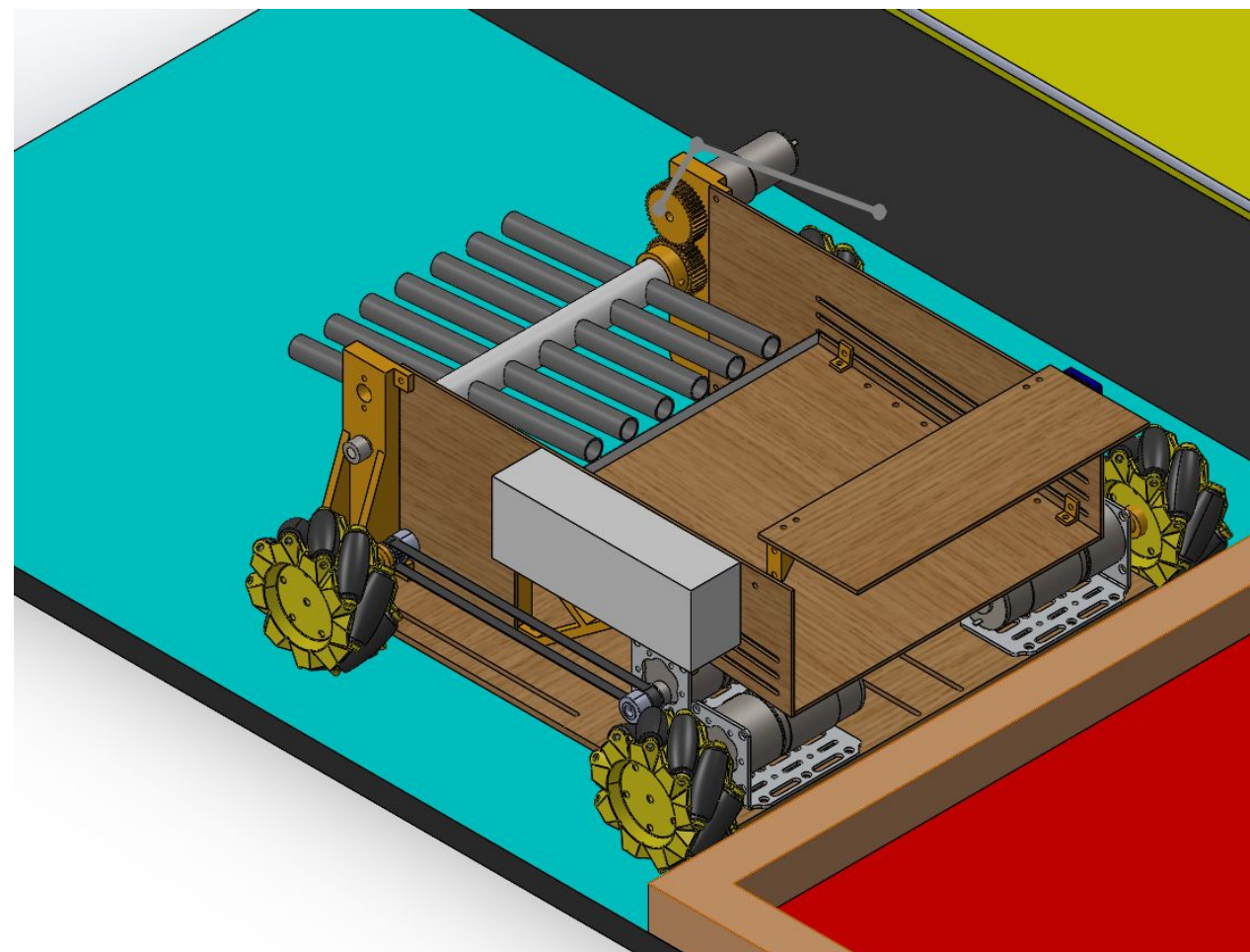
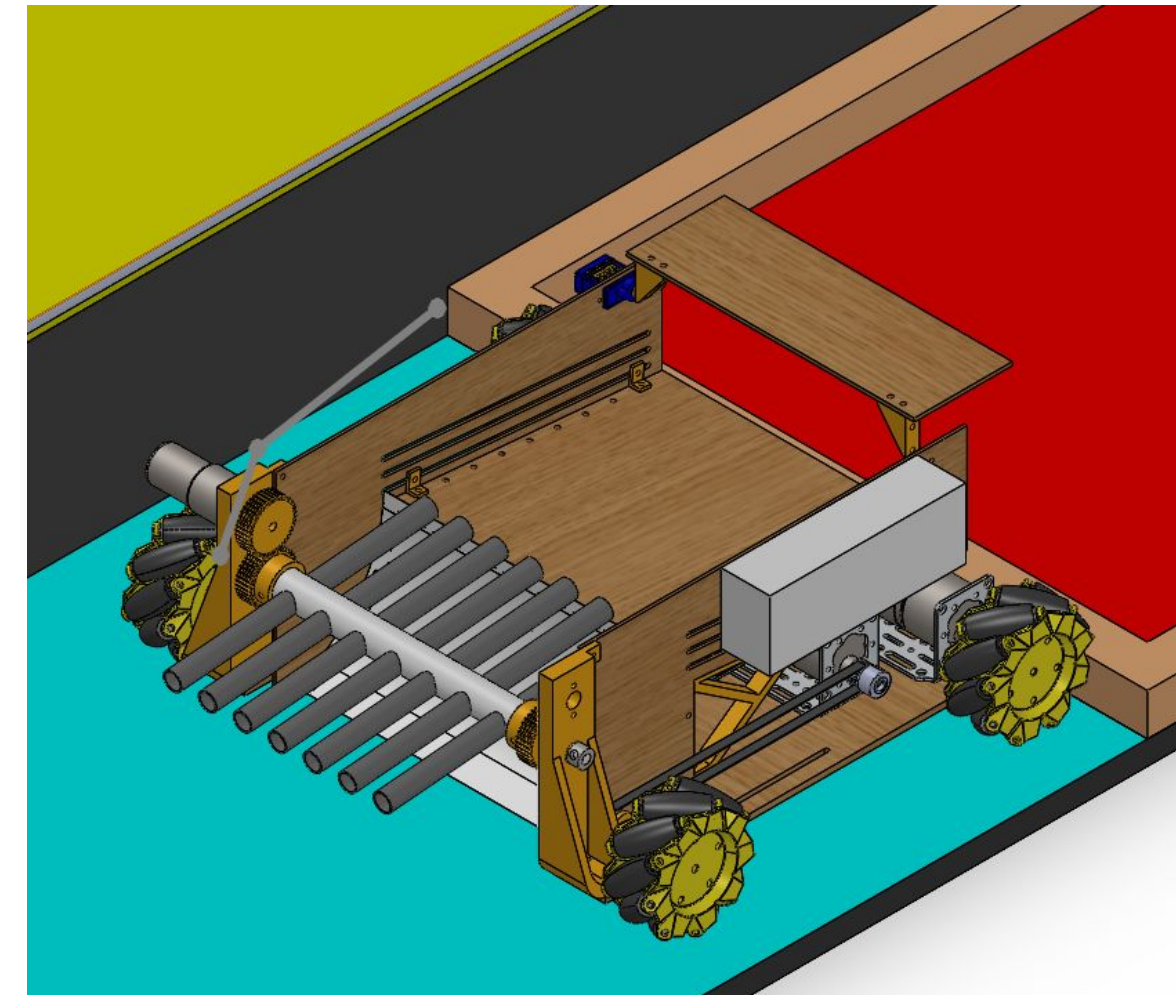
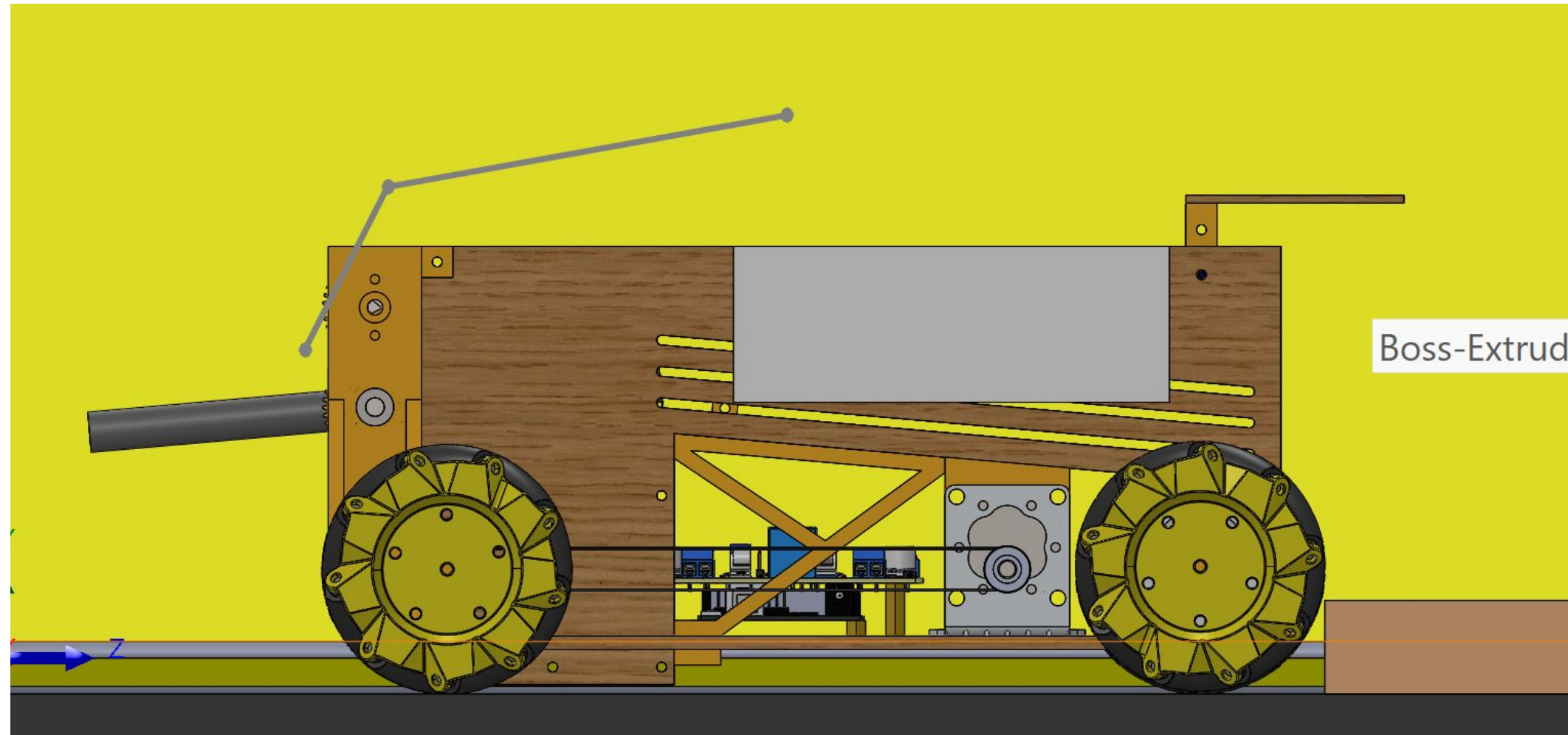
CAD Images (Traverse Seesaw)

PHASE 3



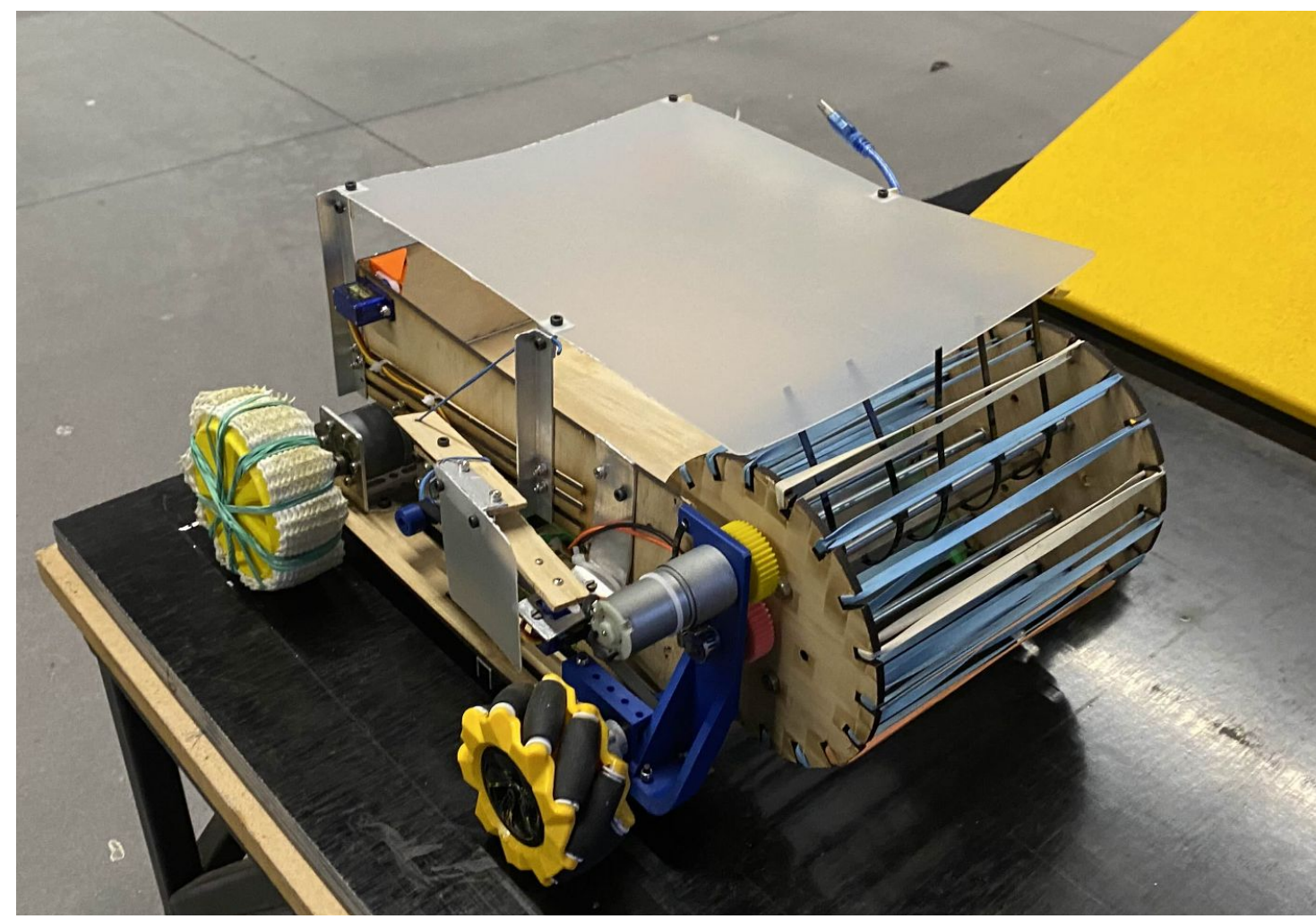
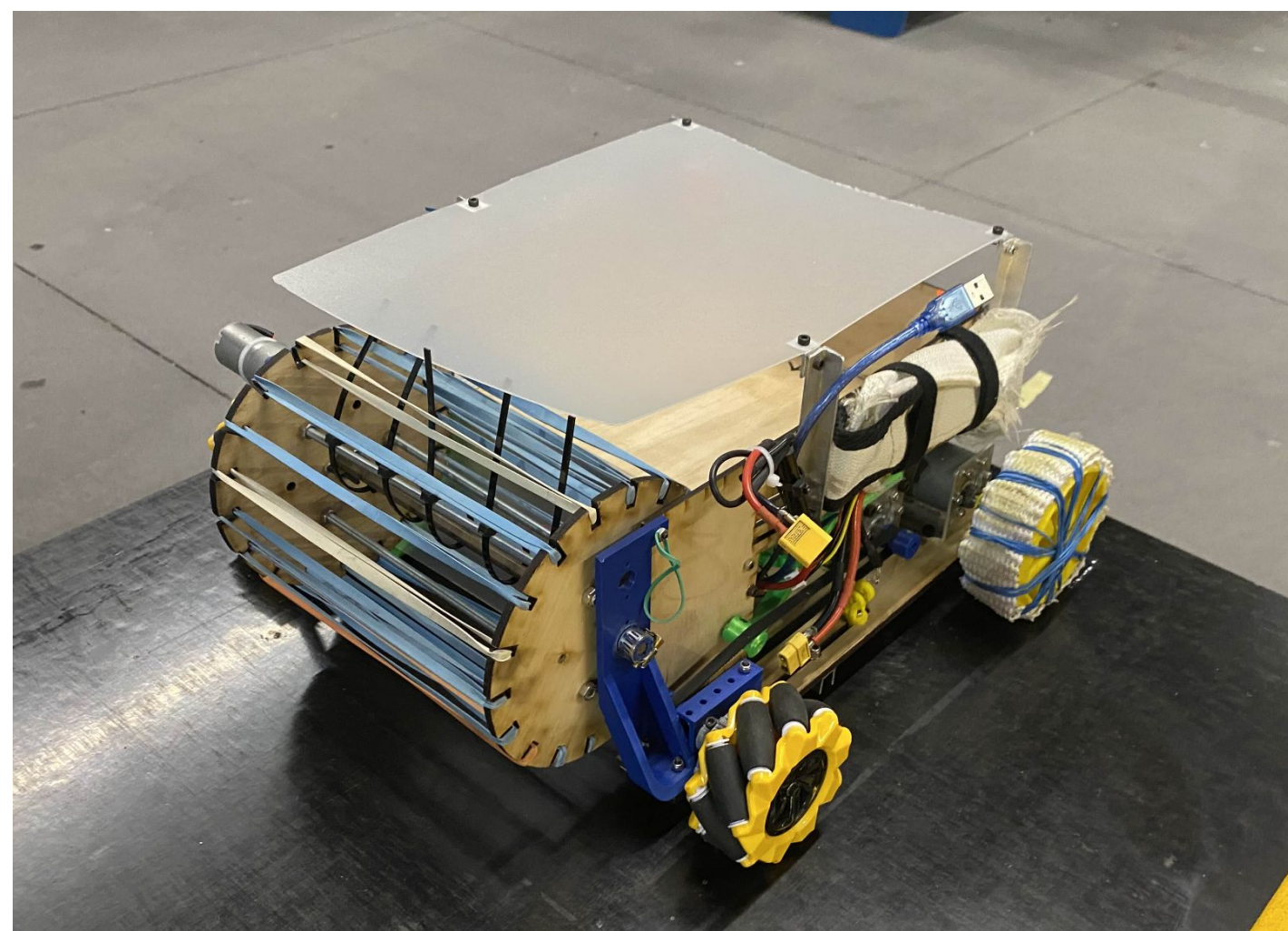
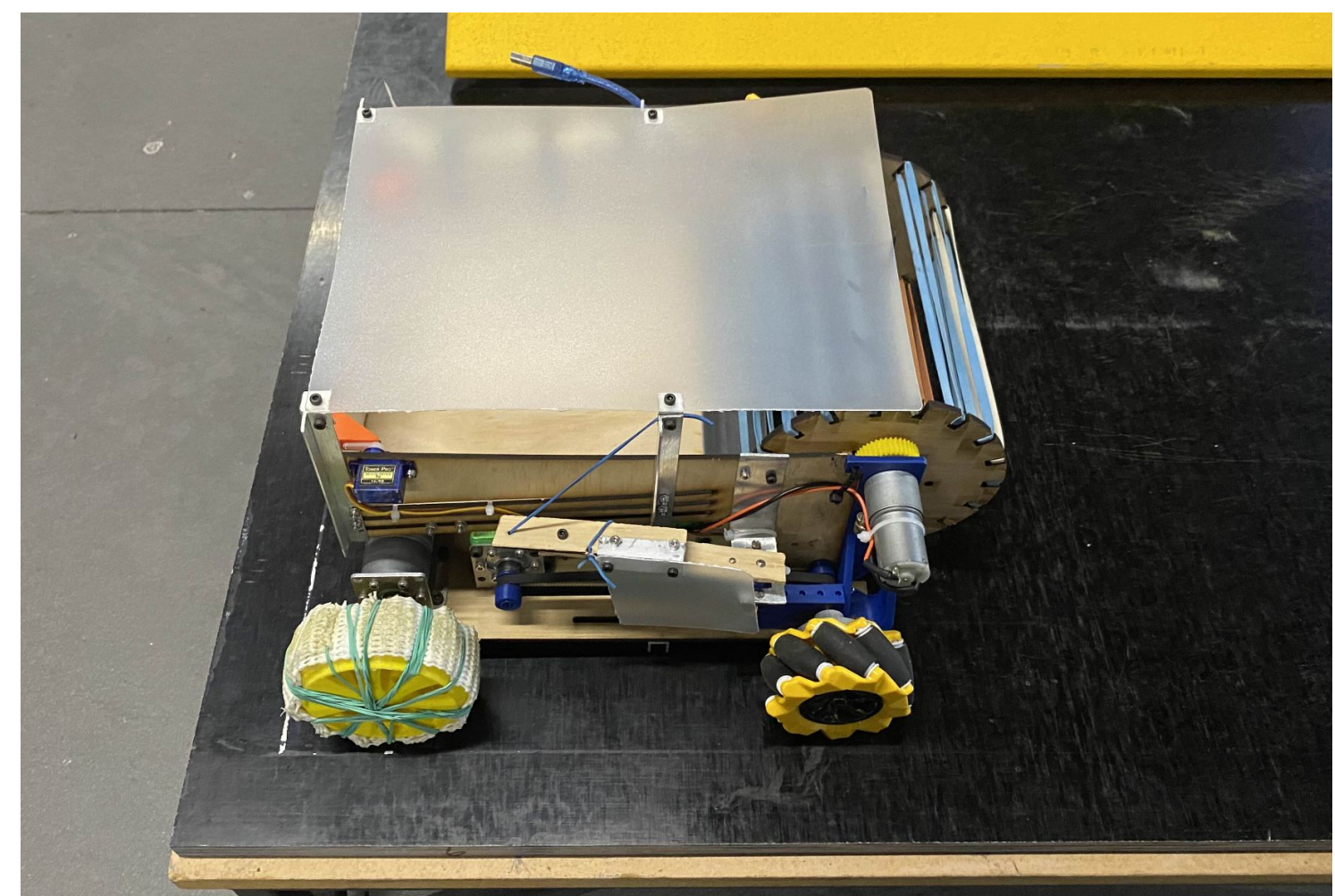
CAD Images (Deposit Ball)

PHASE 4



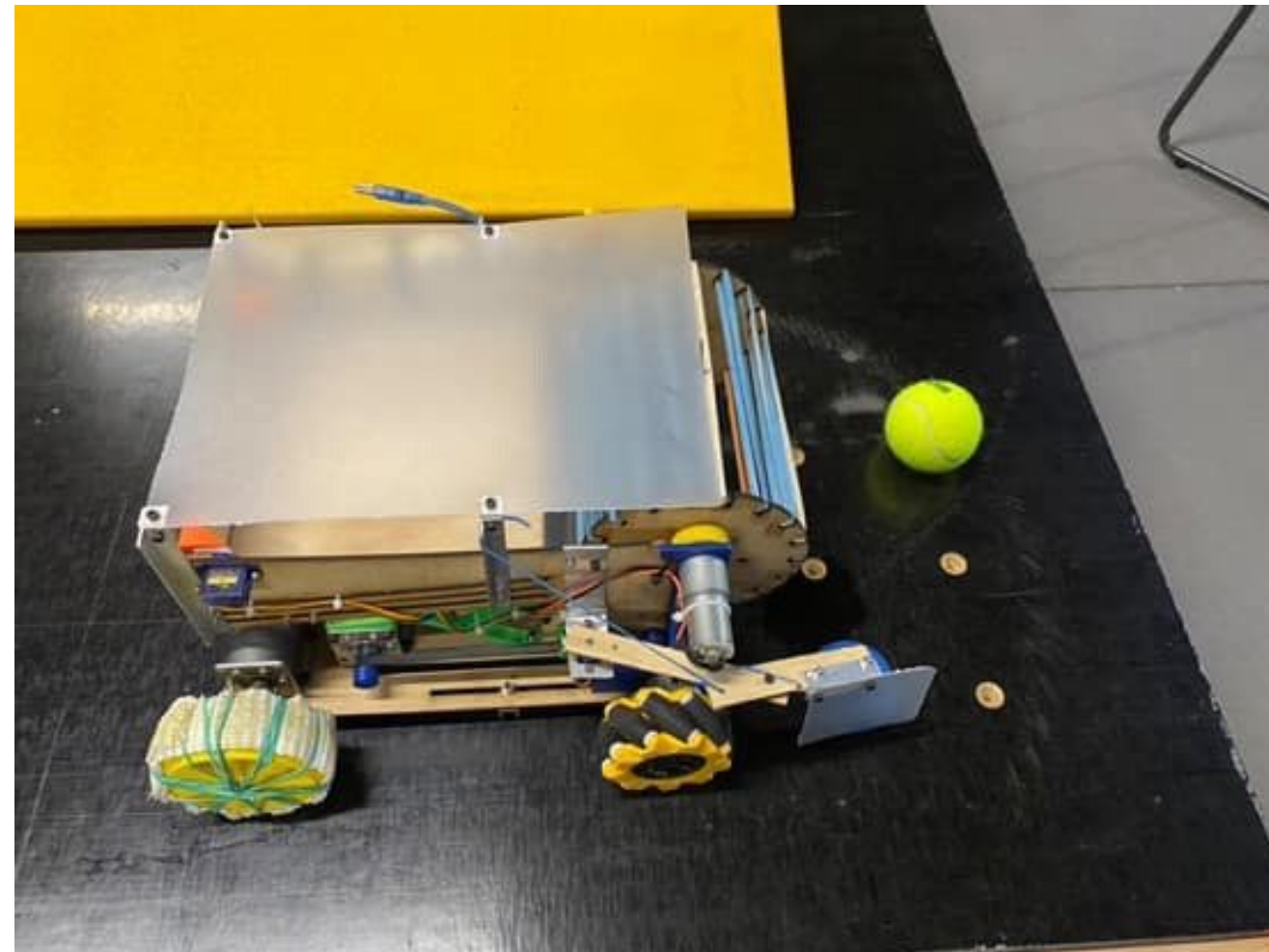
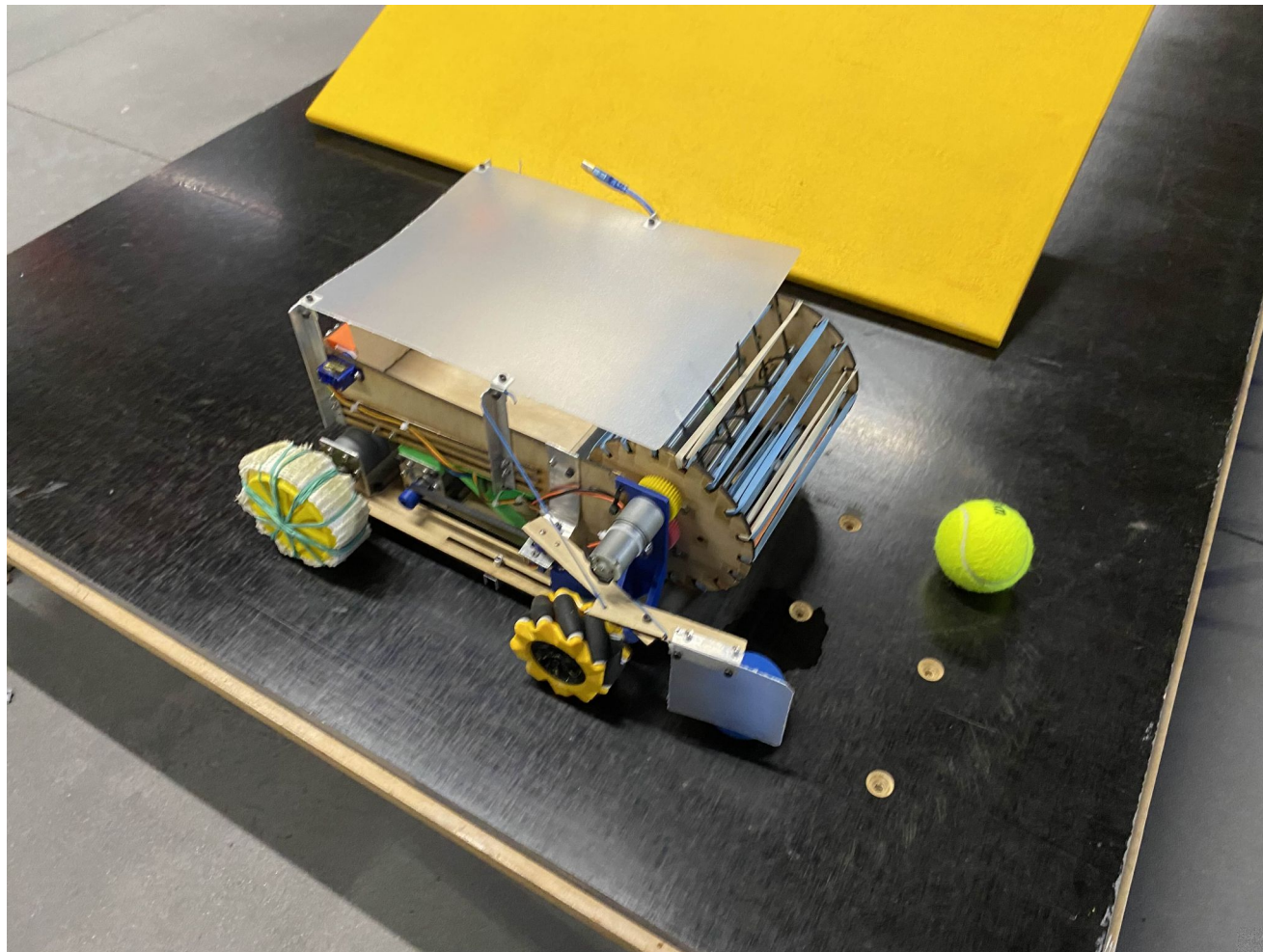
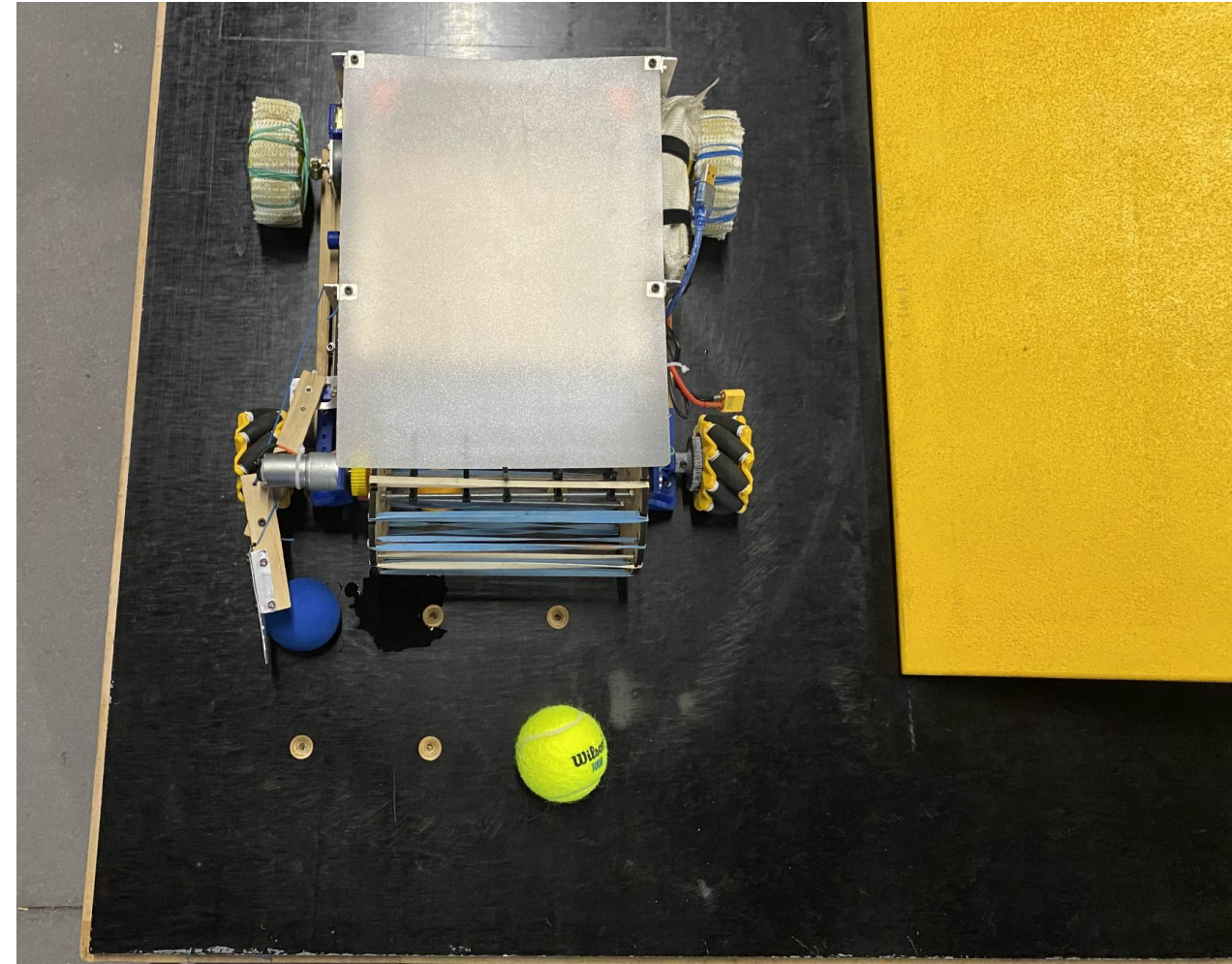
2.2 Final Prototype (System Set-up and ready for start)

PHASE 1



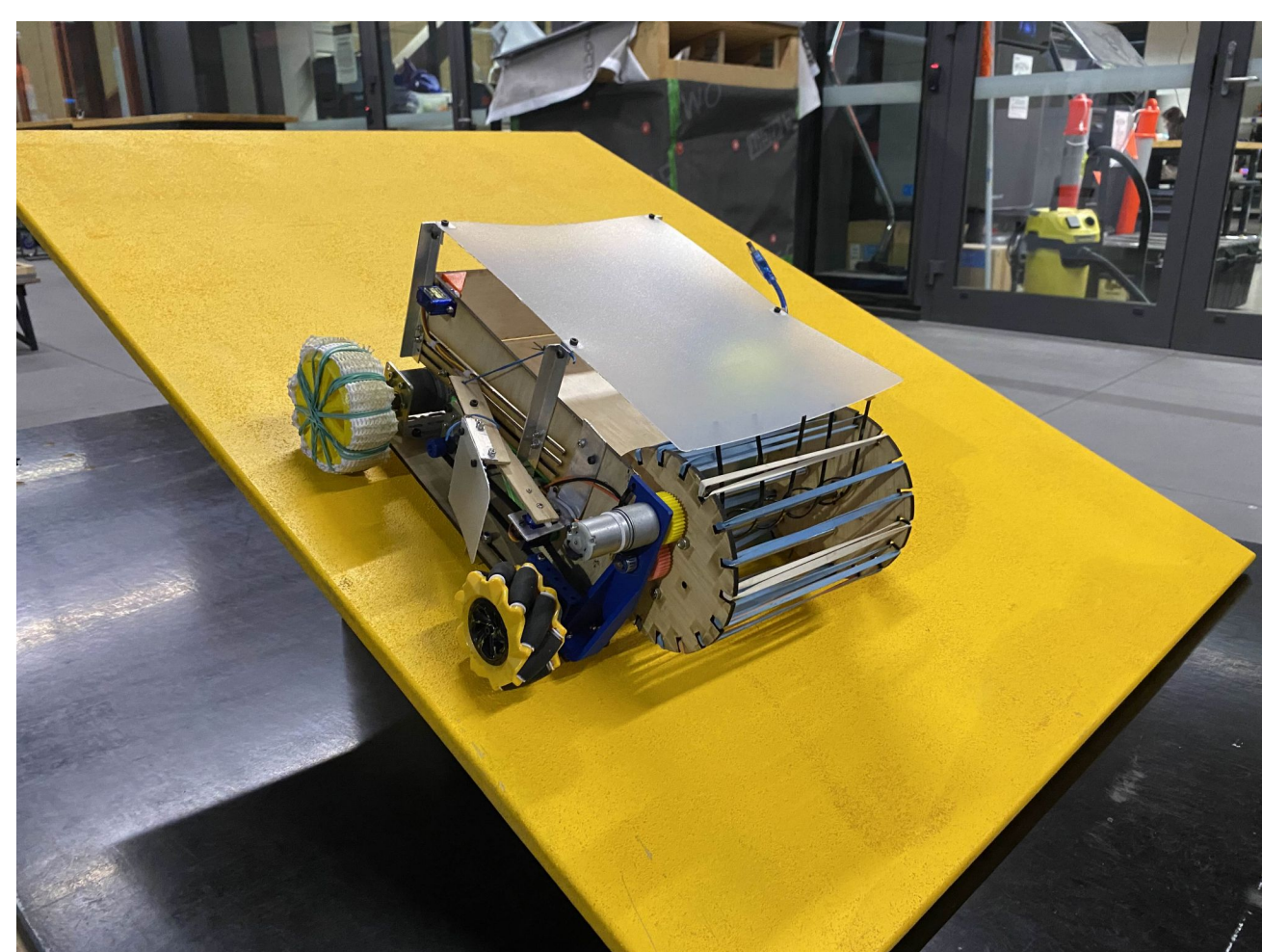
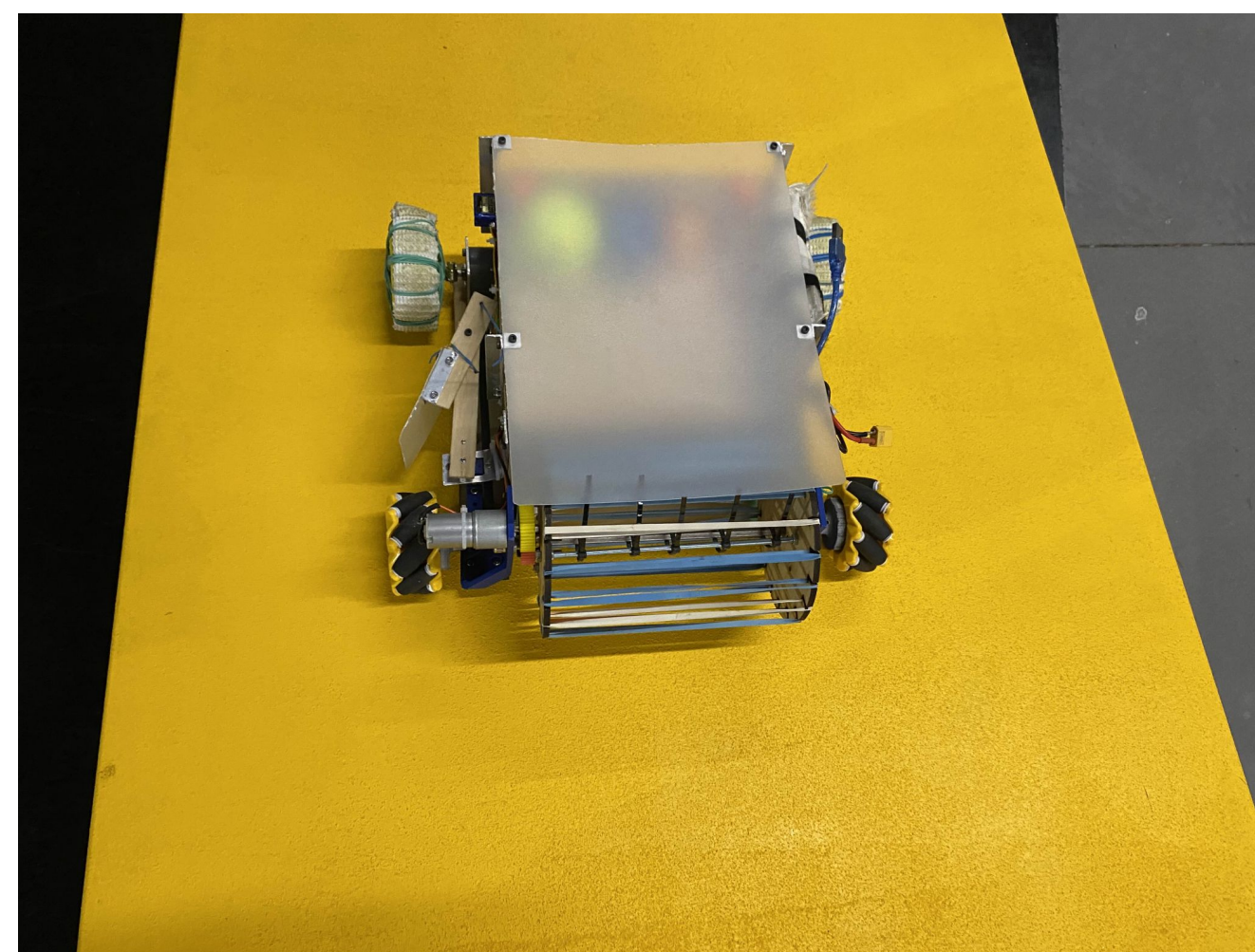
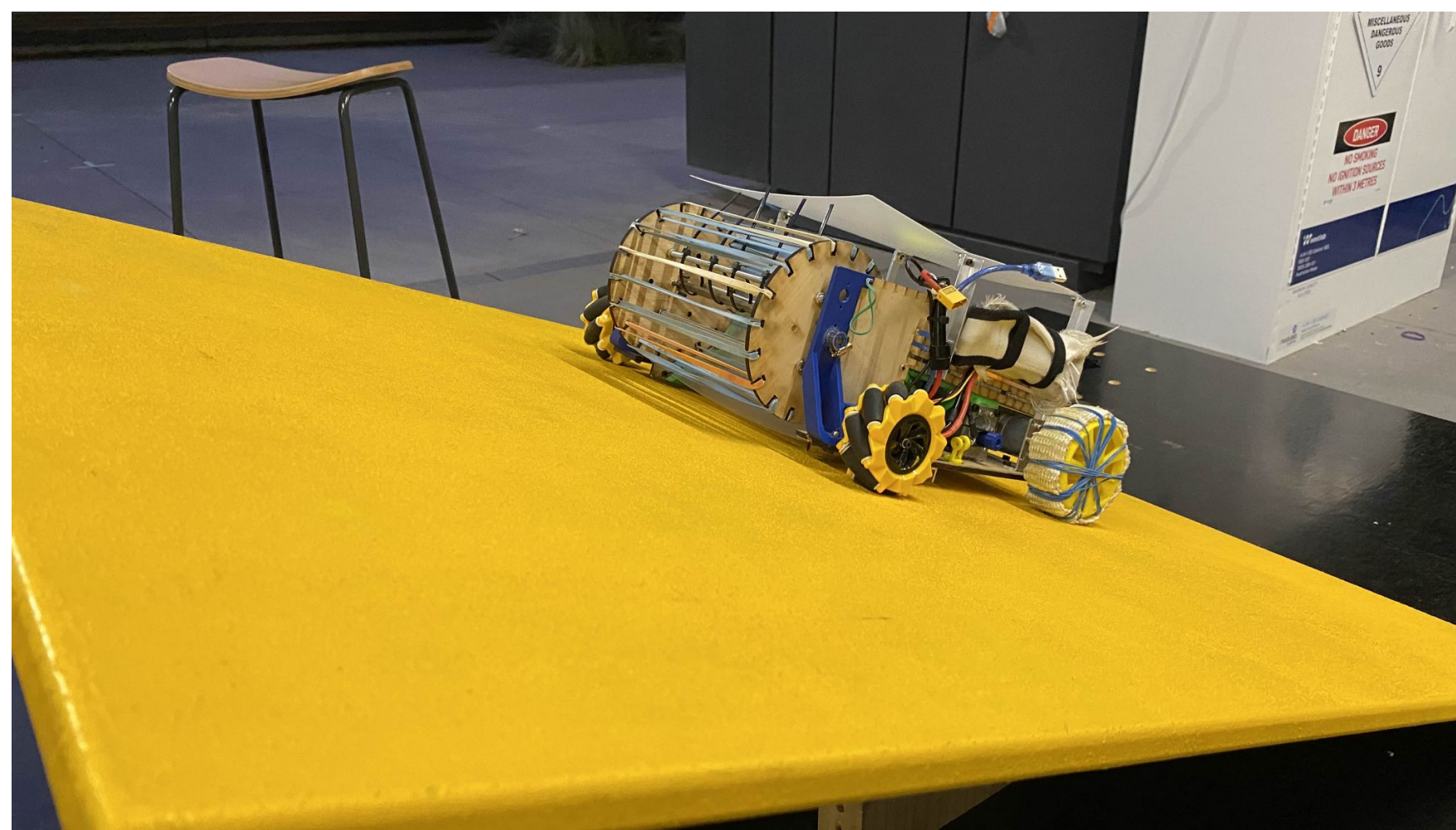
Final Prototype (Pick-Up Ball)

PHASE 2



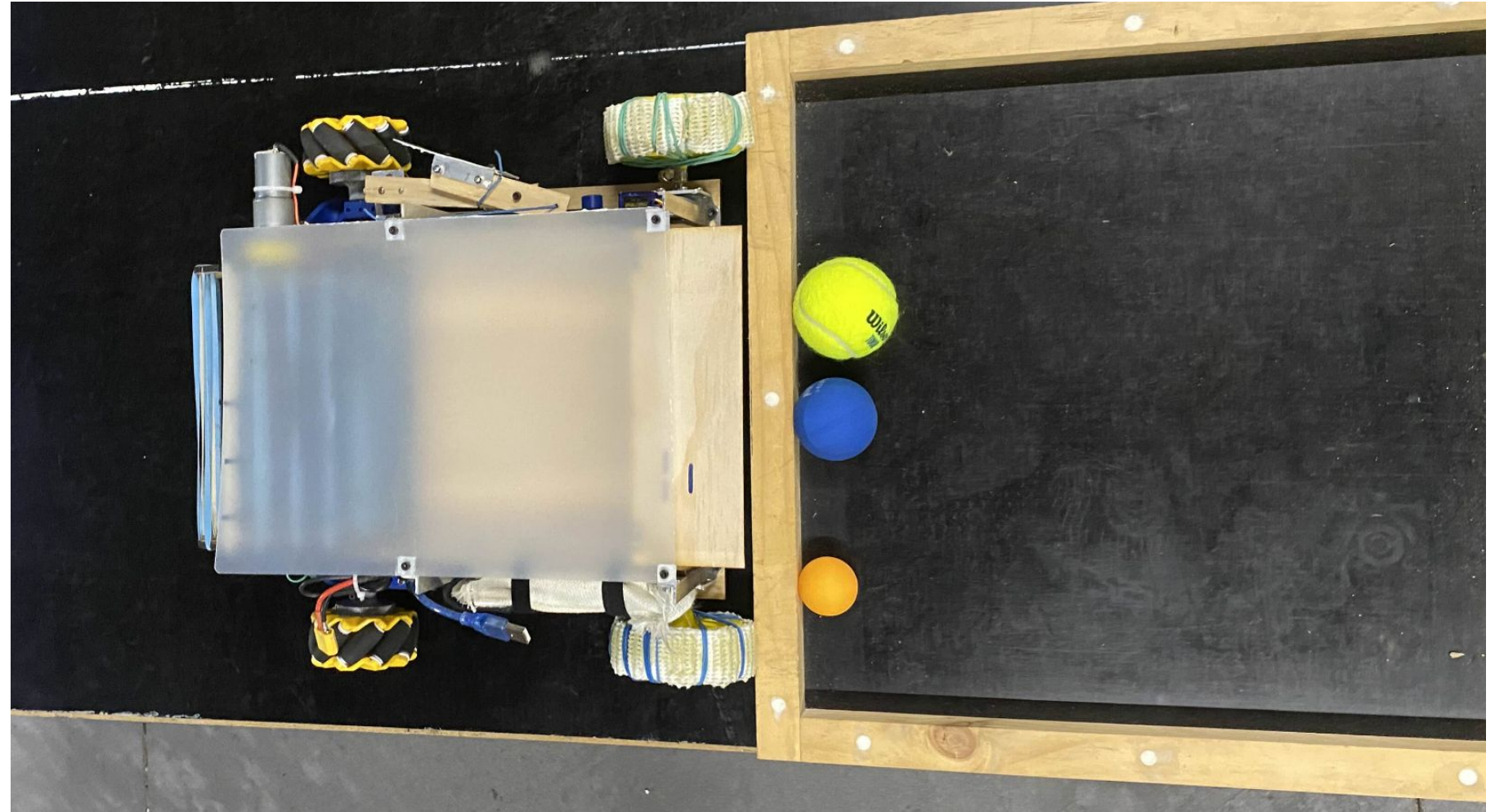
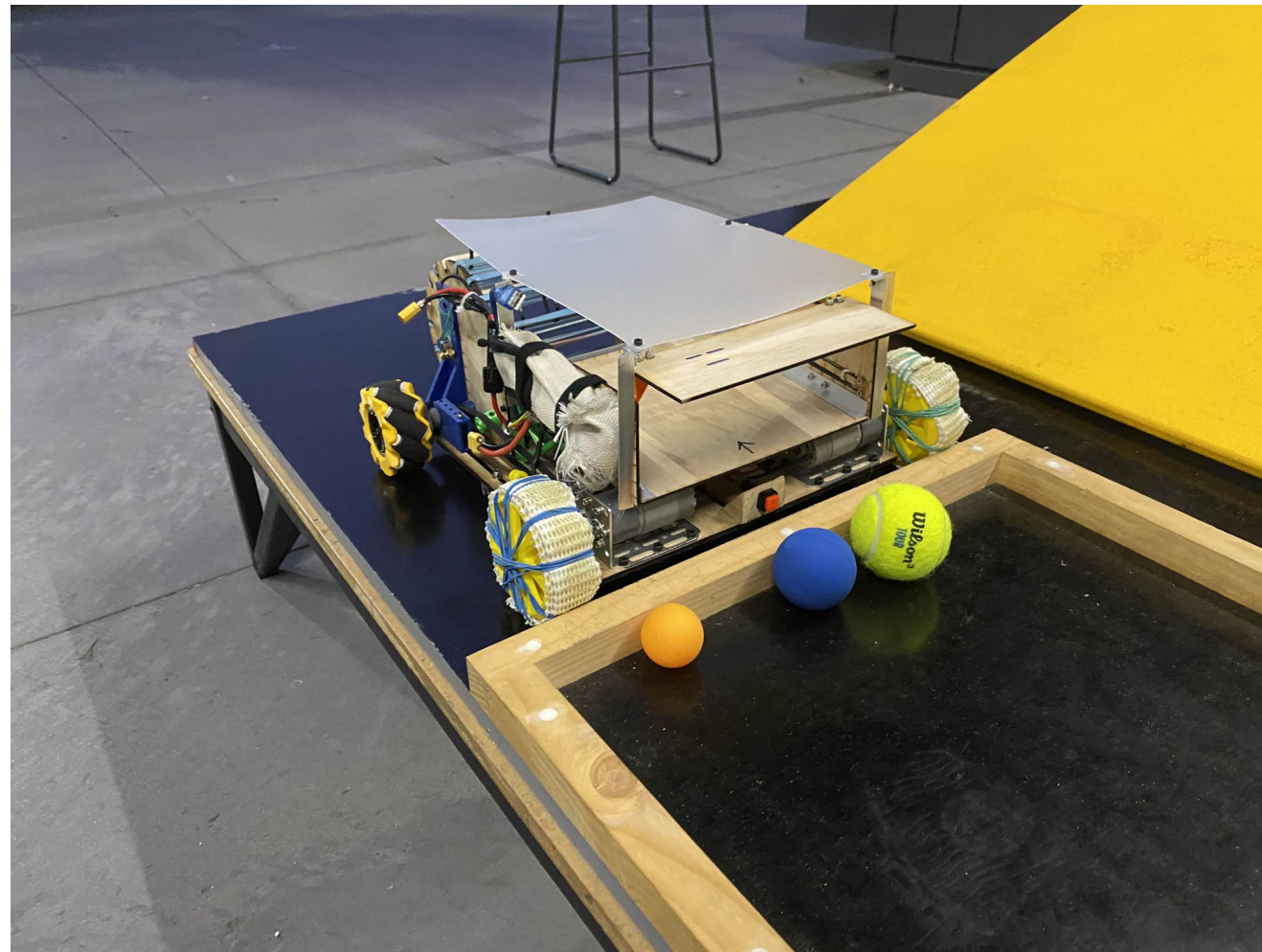
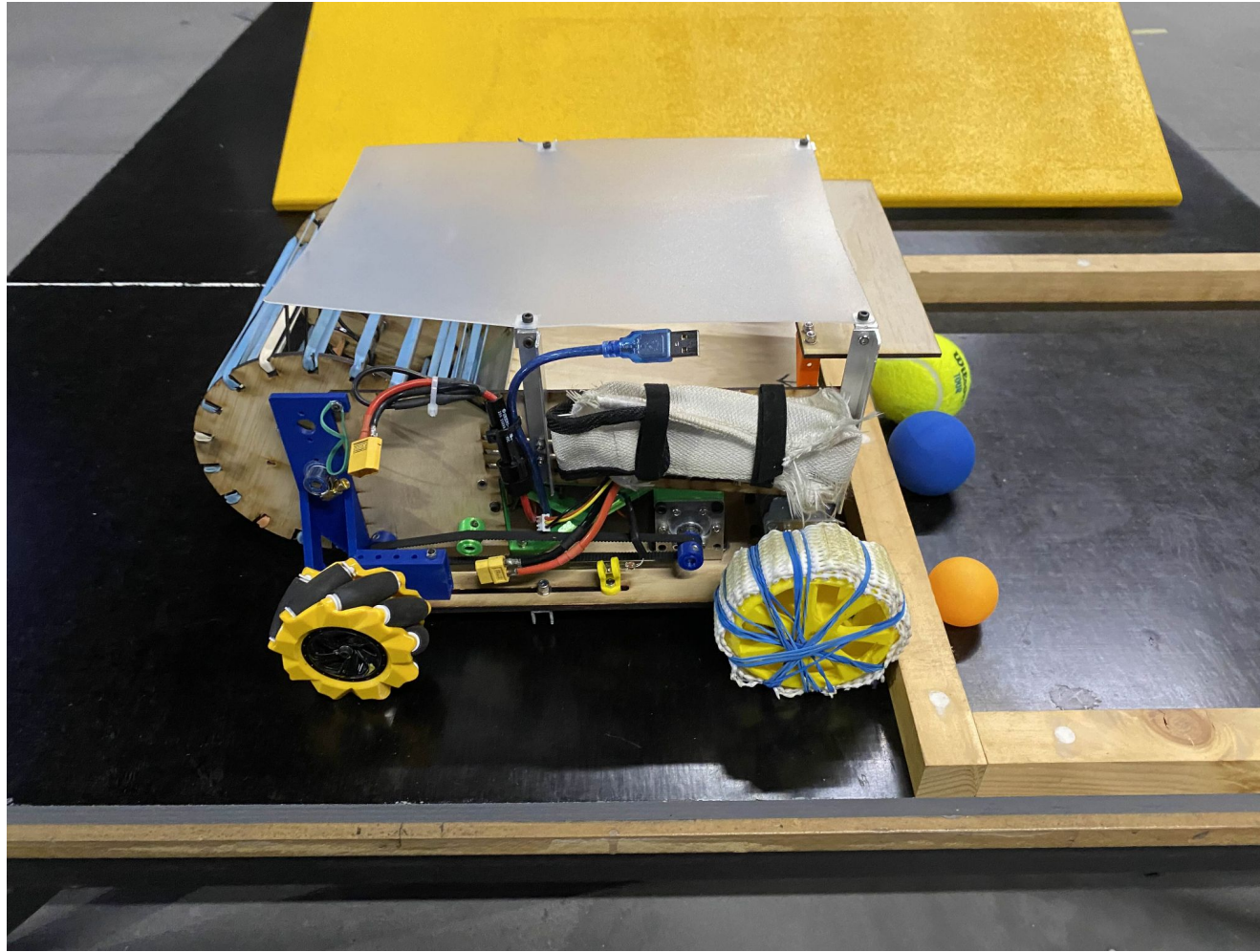
Final Prototype (Traverse Seesaw)

PHASE 3



Final Prototype (Deposit Ball)

PHASE 4



2.3 Design Comparison Details (678 words)

2.3.1 Intake System

In short, our intake system has been changed at least twice because it failed to scoop the balls into our robot. This was due to a false assumption we had made.

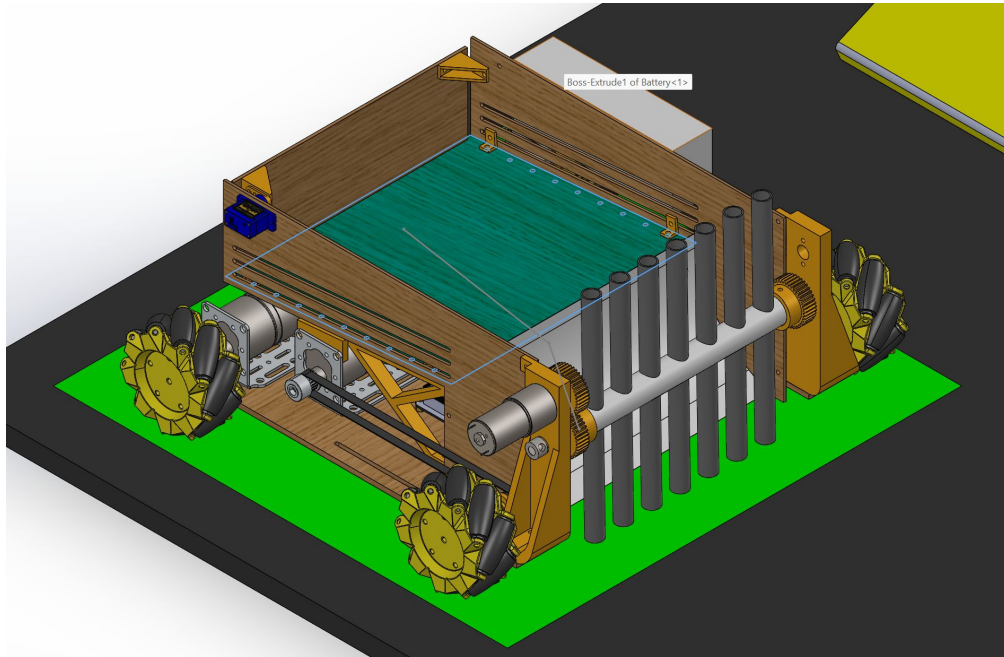


Figure 2.3.1.1
Intake System - Generation 1 CAD
Flex Tube Intake



Figure 2.3.1.2
Intake System - Generation 1
Flex Tube Intake



Figure 2.3.1.3
Intake System - Generation 2
3D-Print blade Intake

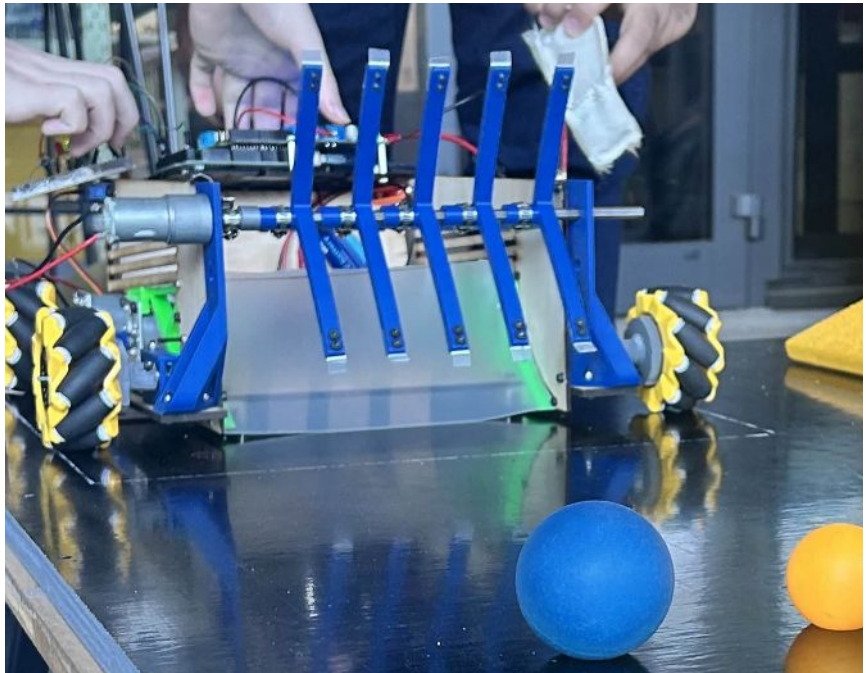


Figure 2.3.1.4
Intake System - Generation 2
3D-Print blade Intake

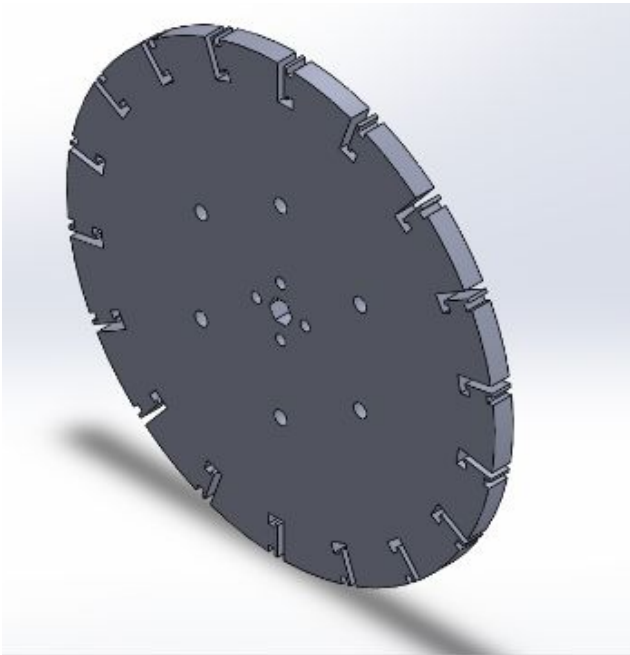


Figure 2.3.1.5
Intake System - Generation 3 CAD
Rubber Band intake

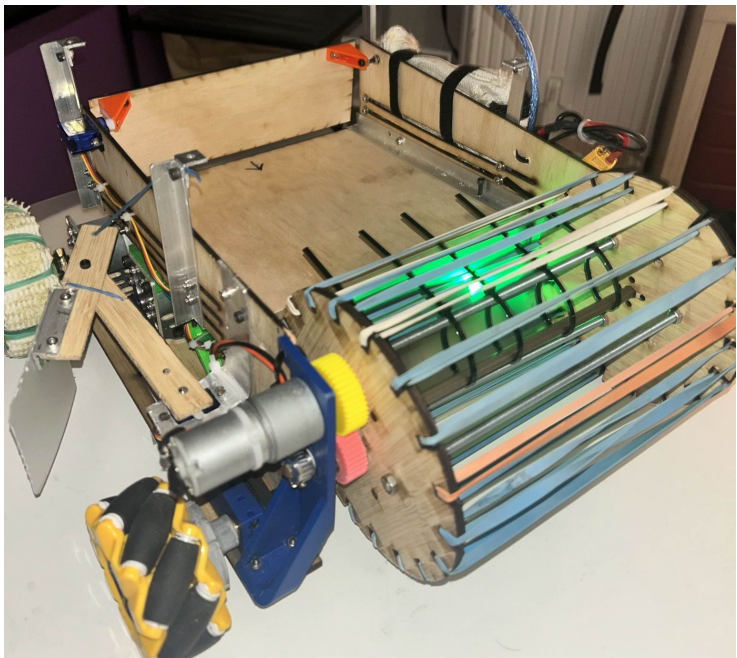


Figure 2.3.1.6
Intake System - Generation 3
Rubber Roller Intake

2.3 Design Comparison Details

2.3.1.1 Generation 1 - Flex Tube Intake

During CAD designing, we assumed that the balls were frictionless and the motor could provide high torque. Therefore, in Generation 1, it was predicted that the flex tube should push the ball into our robot. This system works well for the tennis ball and table tennis, but not for the racquetball.

A little gap exists between the ground and the ramp. For the tennis ball and the table tennis, it is easy to overcome this gap and be pushed into the robot by rubbing the surface of the tube or the ramp (depending on which one has higher friction). However, due to the high surface friction of the racquetball, it got stuck at the gap and the system stopped working. (Figure 2.3.1.1.1)

To solve this problem, we can either increase the torque of the motor so that the friction can be ignored or change the design so that the ball will be scooped up instead of pushed up, which provides an upward force to lift the racquetball (Figure 2.3.1.1.3). Since it is not a budget-friendly option to rebuy a motor with higher torque, we opted for the latter one.

We tried to change the tube to a flex rubber band, wishing that it could deflect if the ball couldn't be pushed up. Sadly, it was rolled up when it was sold and we couldn't twist it straight (Figure 2.3.1.1.5). Also, it had larger friction than the tube, but at least it was a good try.

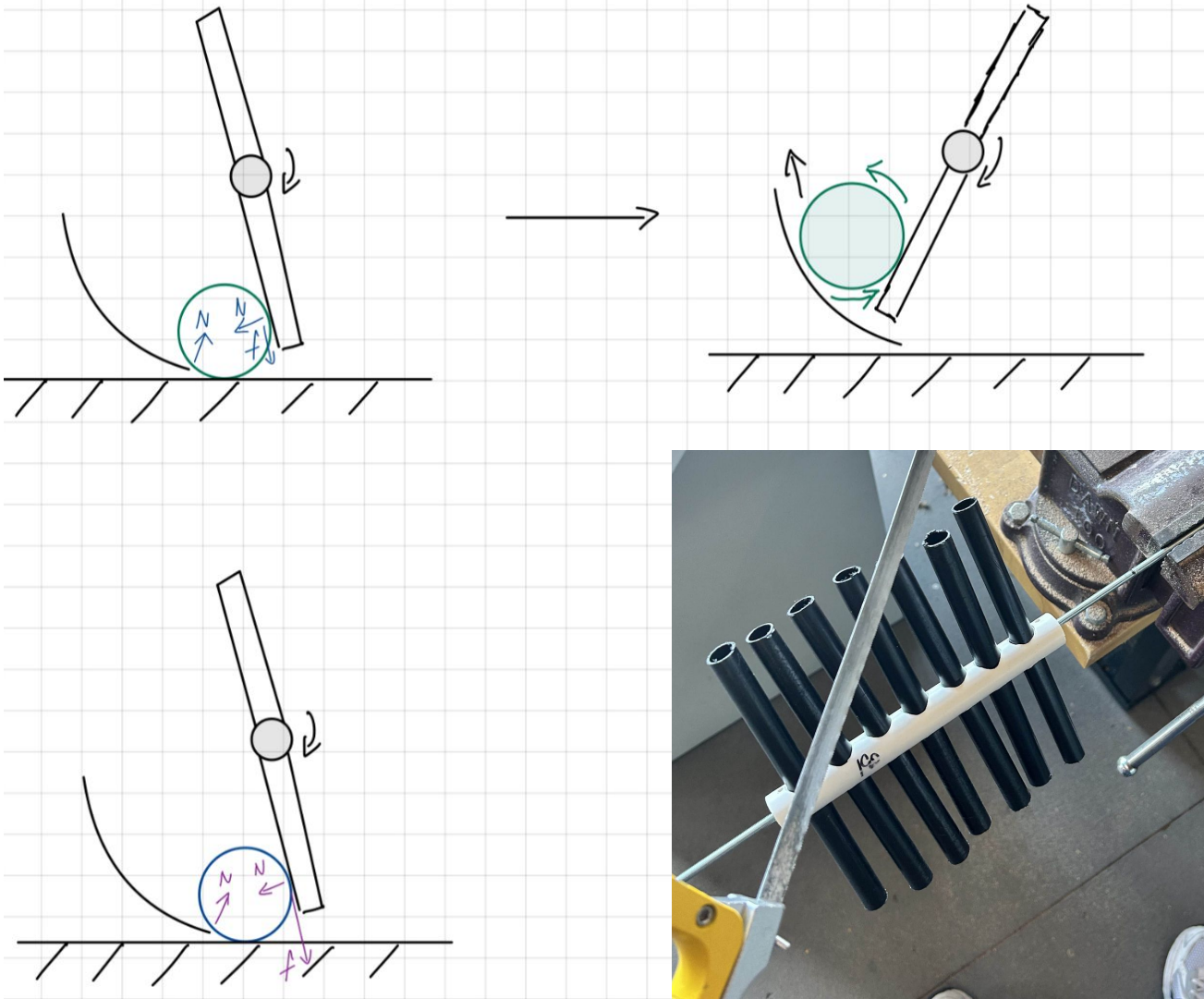


Figure 2.3.1.1.1
Intake Section Draft Drawing



Figure 2.3.1.1.2
Generation 1 - Flex Tube Intake

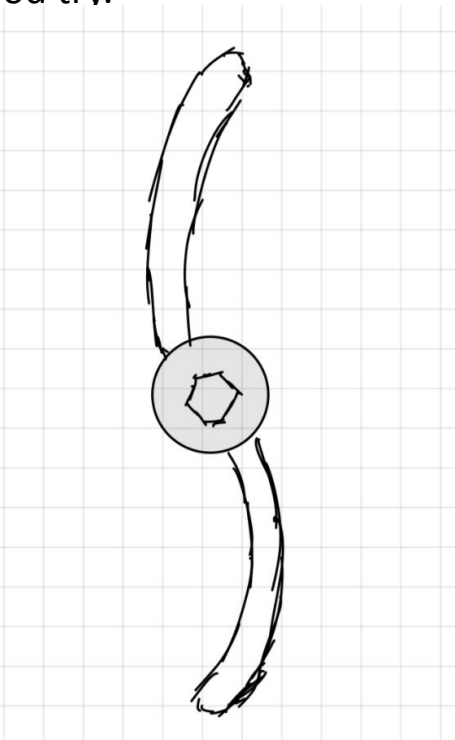


Figure 2.3.1.1.3
Generation 2 - Draft Drawing

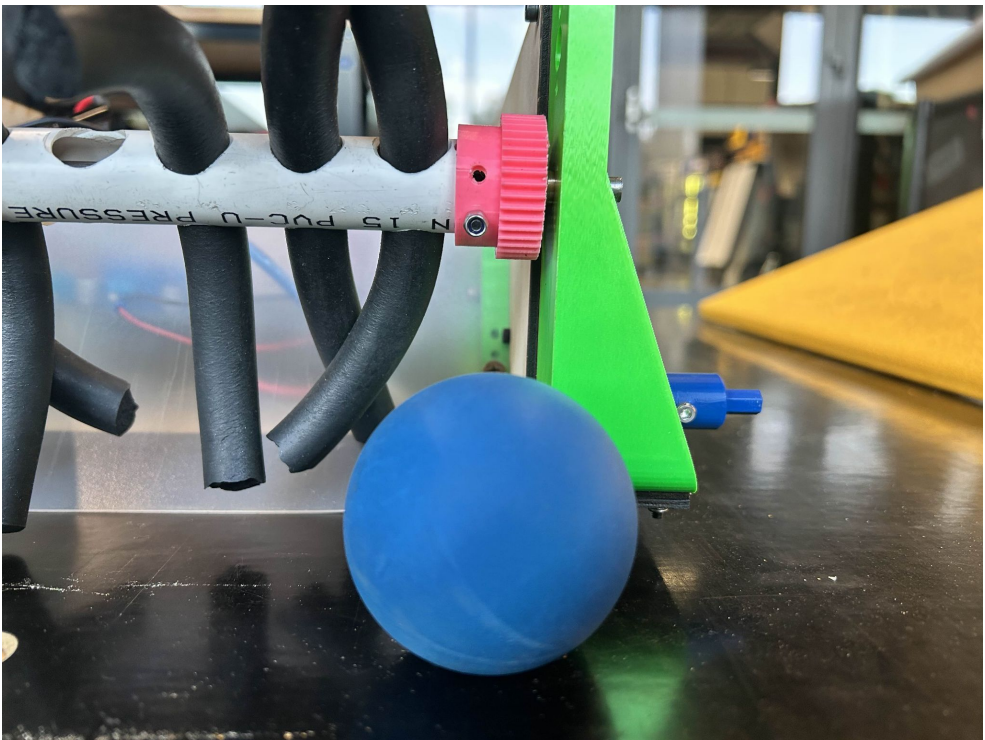
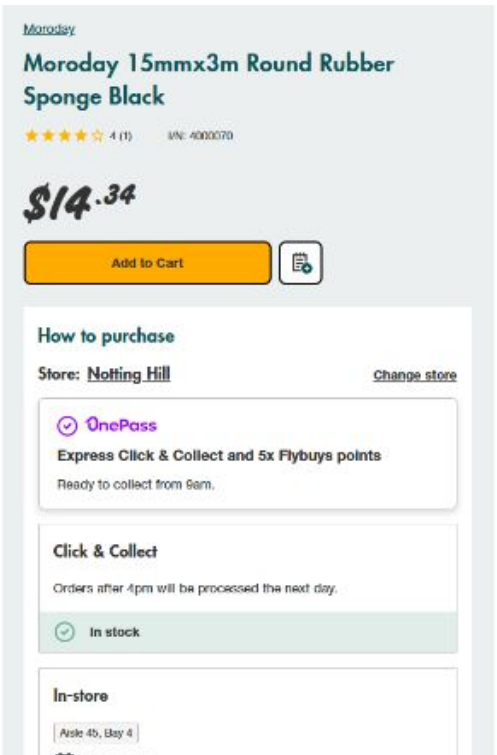


Figure 2.3.1.1.4
Generation 1 - Rubber Bank Intake



Figure 2.3.1.1.5
Rubber Band



2.3 Design Comparison Details

2.3.1.2 Generation 2 - 3D-Print Blade Intake

Instead of using a straight blade to push the balls, we designed a curved blade so that it could provide lift force to the balls at the bottom during intake. Thin plastic sheets were attached to the tiny gaps at the end of the blade to provide flexibility while it was approaching the balls.

Although it provided some lift, the problem still existed as the curvature was not large enough. Furthermore, the robot “jumped” when the blade approached the balls at a certain position due to the rigidity of PLA (Figure 2.3.1.2.3). Moreover, smaller balls like table tennis balls were thrown out after they reached the top of the ramp.

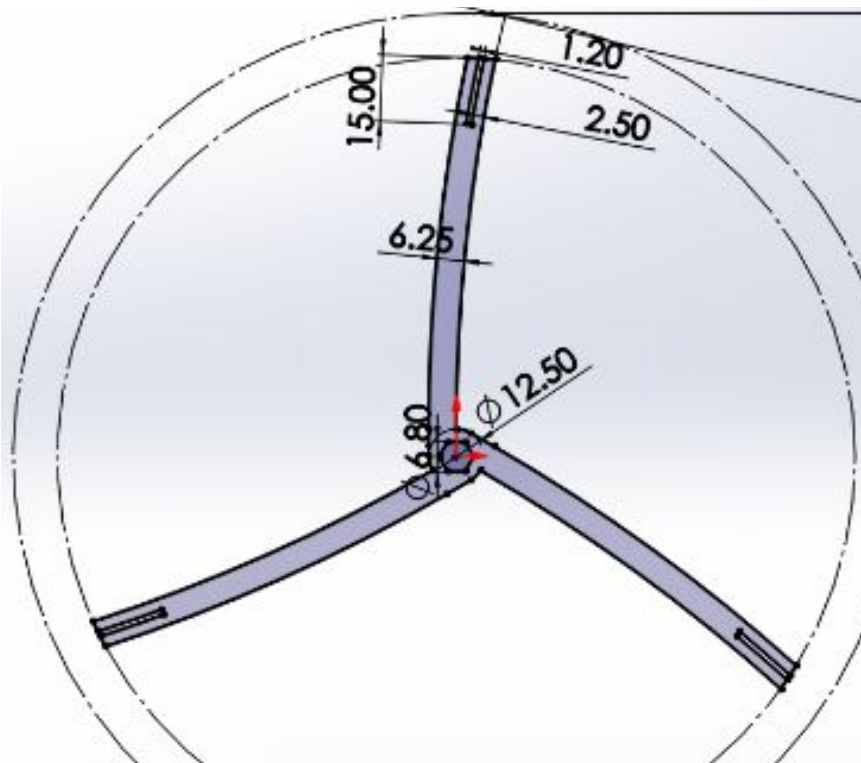


Figure 2.3.1.2.1
Generation 2 - CAD

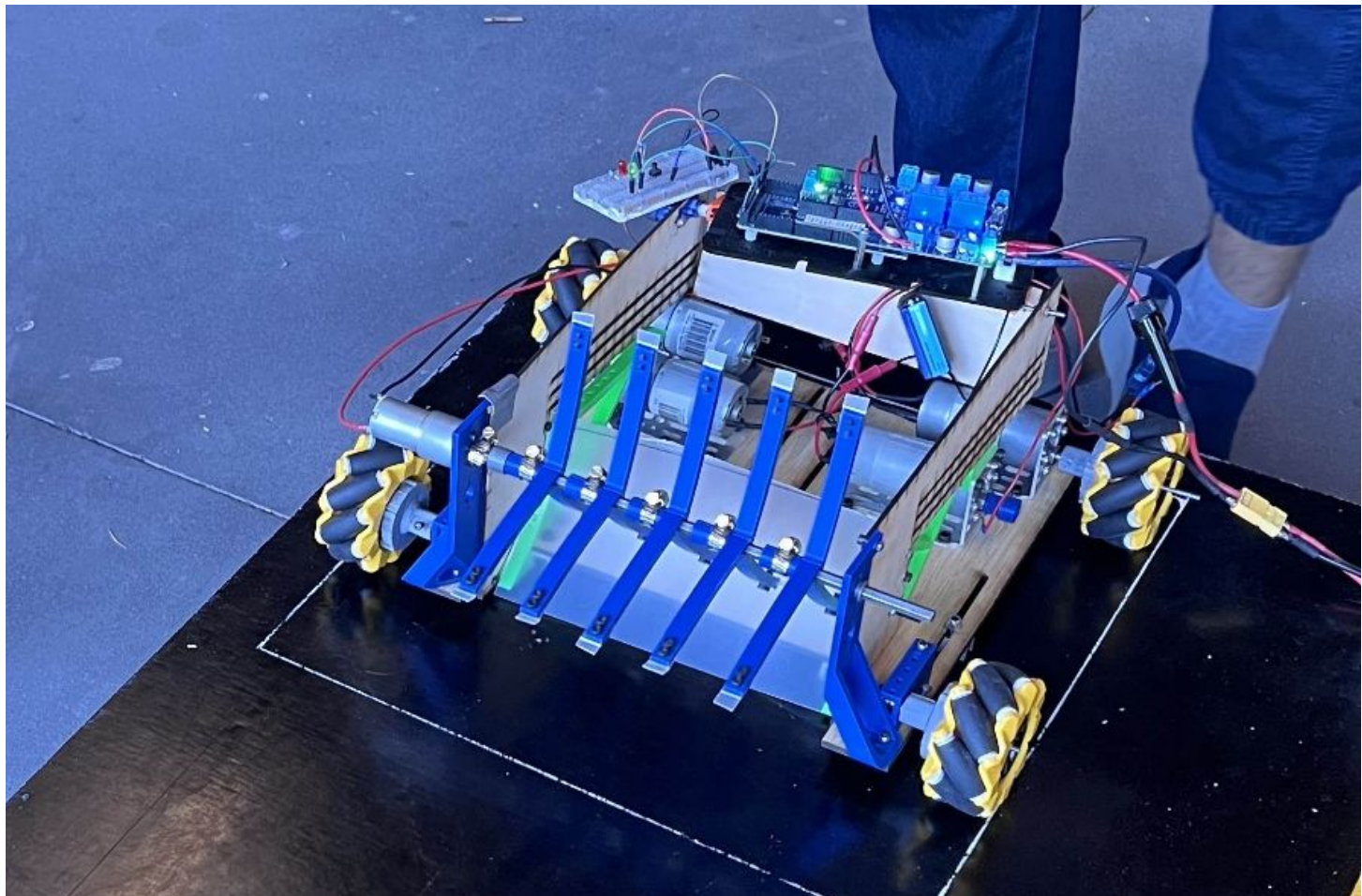


Figure 2.3.1.2.2
Generation 2 - 3D-Print Intake

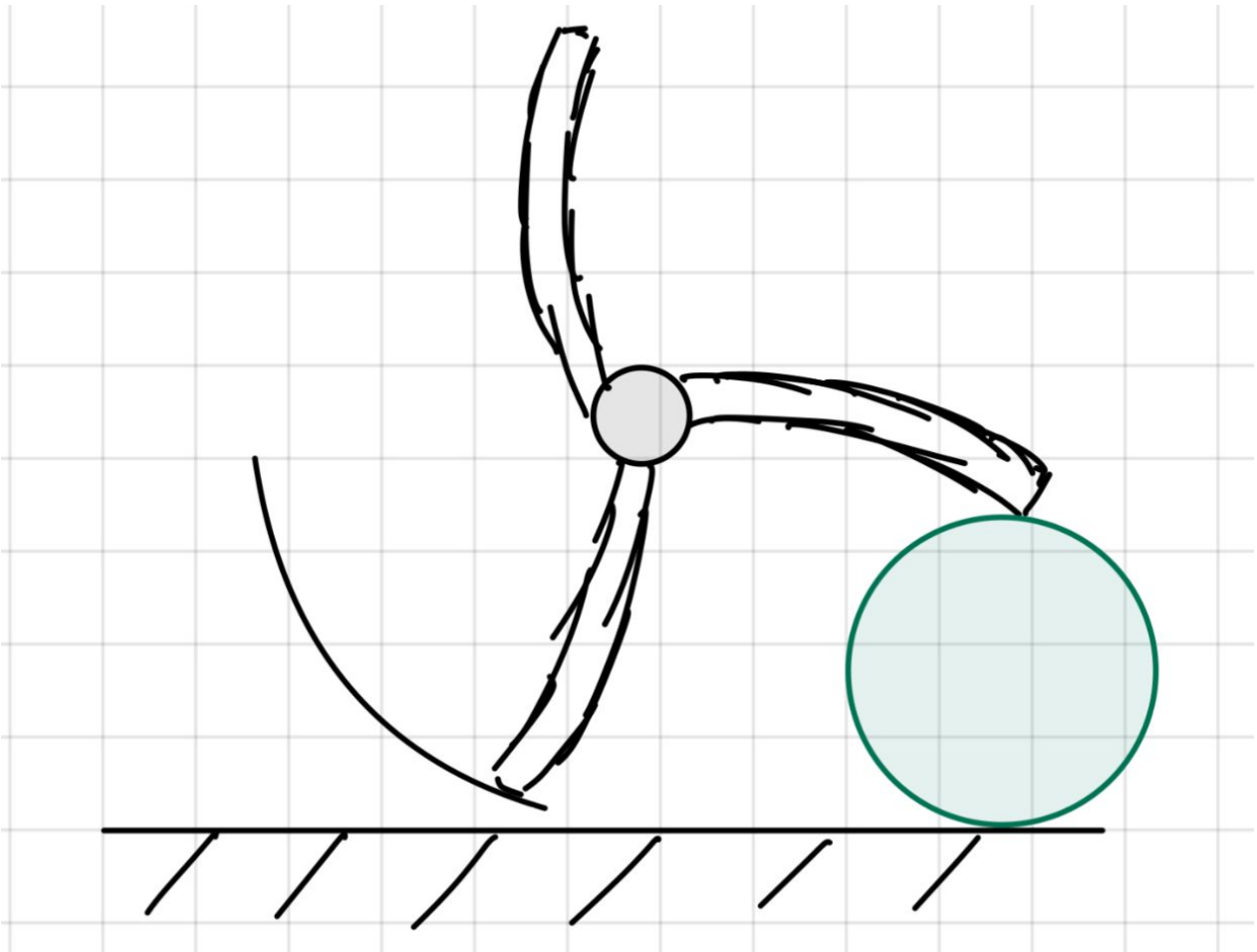


Figure 2.3.1.2.3
Generation 2 - Stuck at certain angle

2.3 Design Comparison Details

2.3.1.3 Generation 3 - Rubber Roller Intake

This is our final intake design. It consists of 2 roller plates and 3 threaded rods to adjust the distance between the 2 plates, tightened with rubber bands. When it approached different balls, the rubber bands compressed to the corresponding size and pushed the ball up by friction. We also added some zip ties to push the table tennis ball up in case the rollers couldn't roll it in.

We noticed that the tennis ball might be launched due to the high elastic energy stored in the rubber bands. Therefore, we added a lid on top.

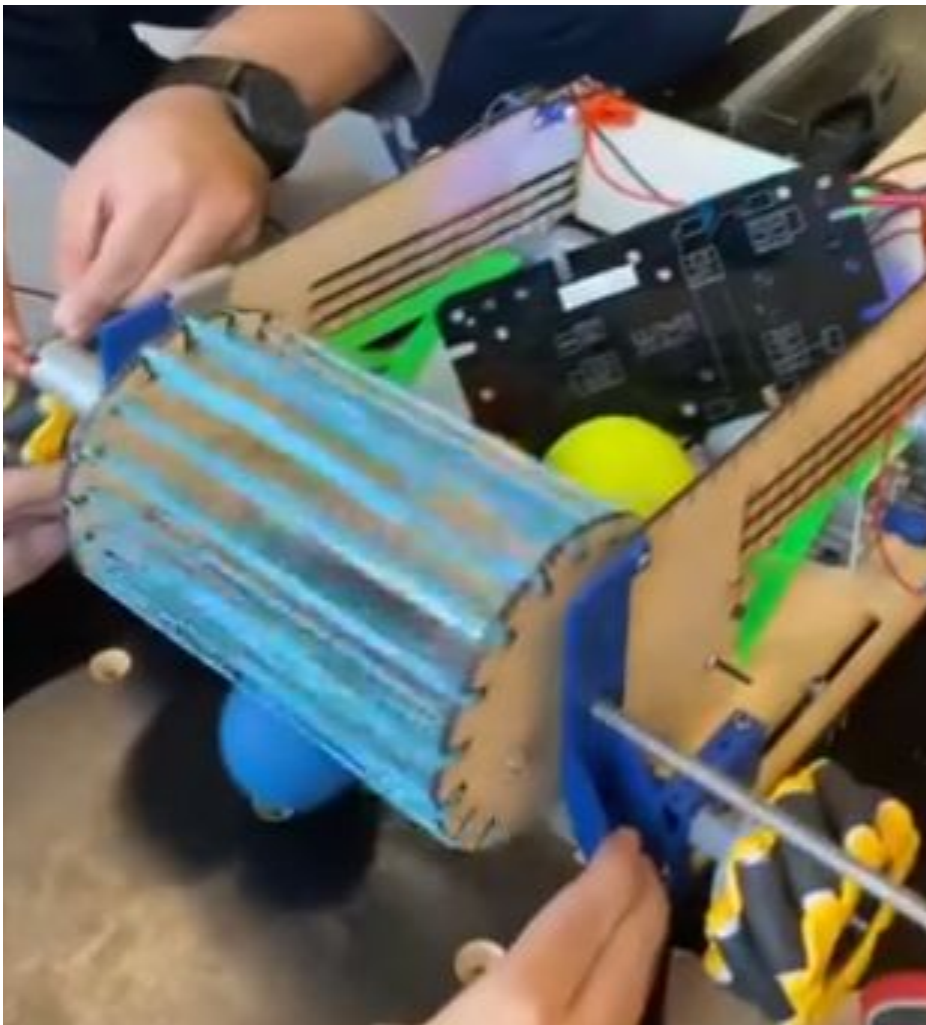


Figure 2.3.1.3.1
Generation 3 - Rubber Roller Intake

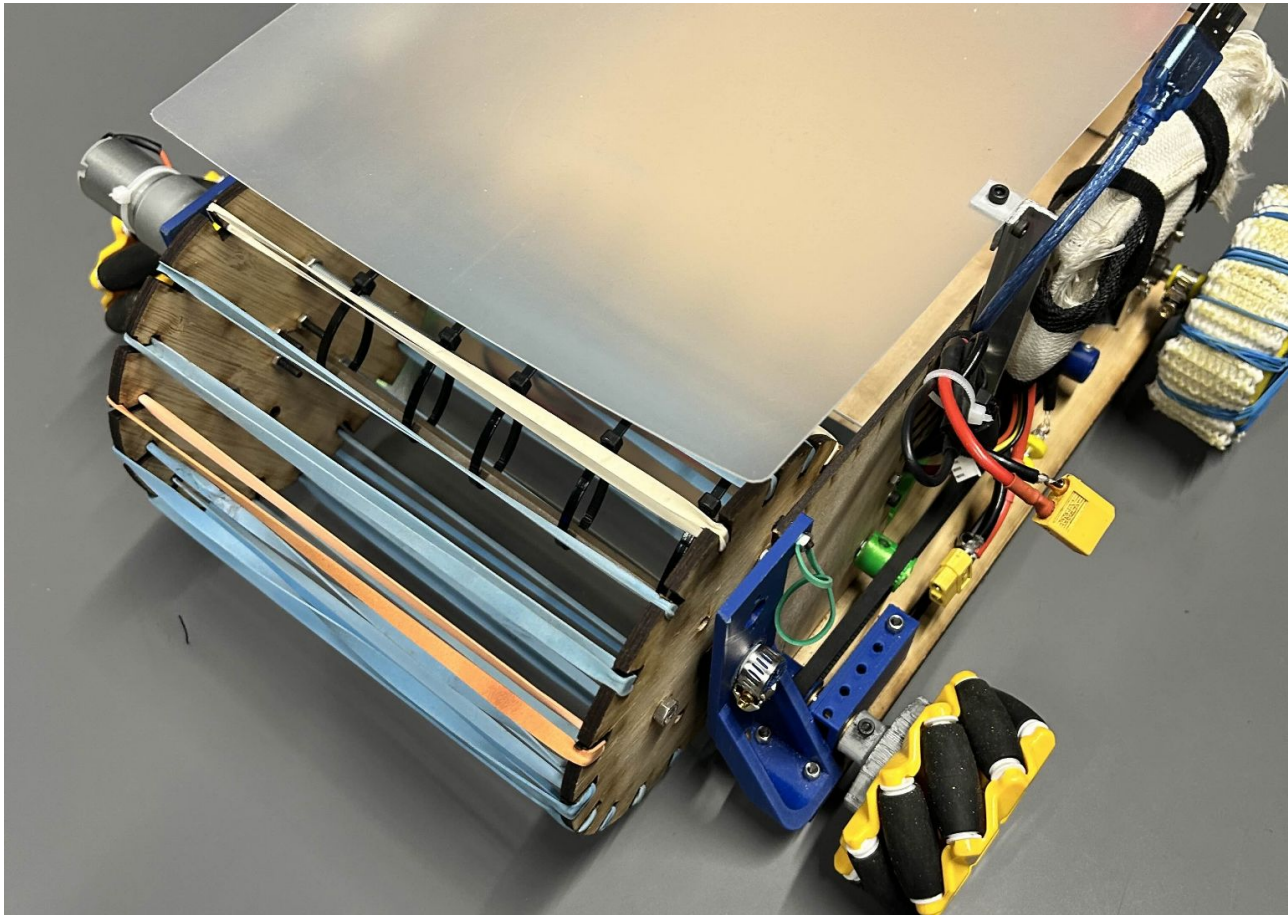


Figure 2.3.1.3.4
Robot with lid

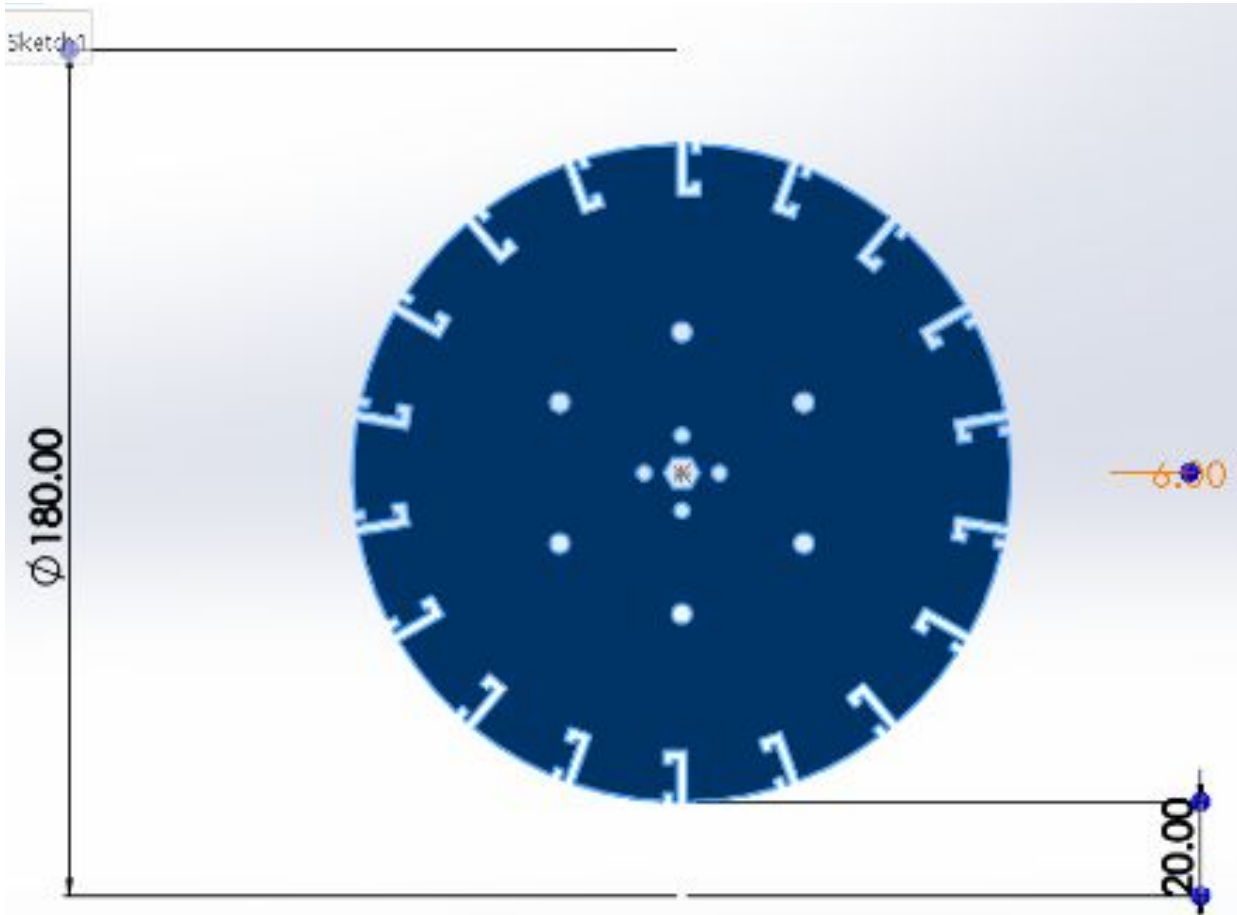


Figure 2.3.1.3.3
Roller Plate CAD

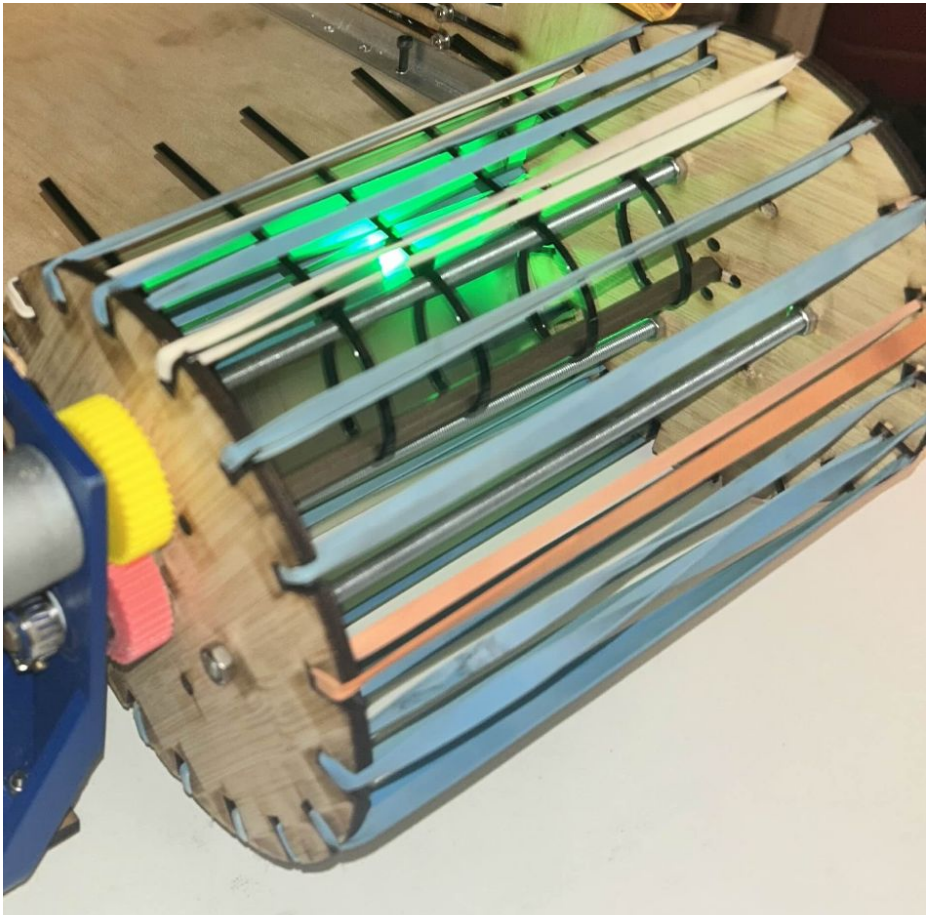


Figure 2.3.1.3.2
Zip ties

2.3 Design Comparison Details

2.3.2 Rear Wheels

In our original design, we planned to use a mecanum wheel chassis. However, the rear wheels got stuck at the seesaw (Figure 2.3.2.2). This was due to a few reasons:

- The assumption of the seesaw as an inclined slope from the field, ignoring the thickness of the seesaw
- Not enough grip from the wheels

As a result, we needed to increase the grip of the wheels and also make something like snow chains.

We 3D printed wheels with a flat surface so that we could try different materials on them. Firstly, we tried double-sided tape with rubber bands acting like snow chains. It worked well, but only for a day, as it lost its stickiness after some trials. Also, it was expensive.

In the final prototype, we used 3 layers of non-slip rug and rubber bands (Figure 2.3.2.5). The rug deformed when the wheel approached the seesaw, providing high friction, while the rubber bands acted like snow chains to climb up the seesaw. Most importantly, the rug was cheap.

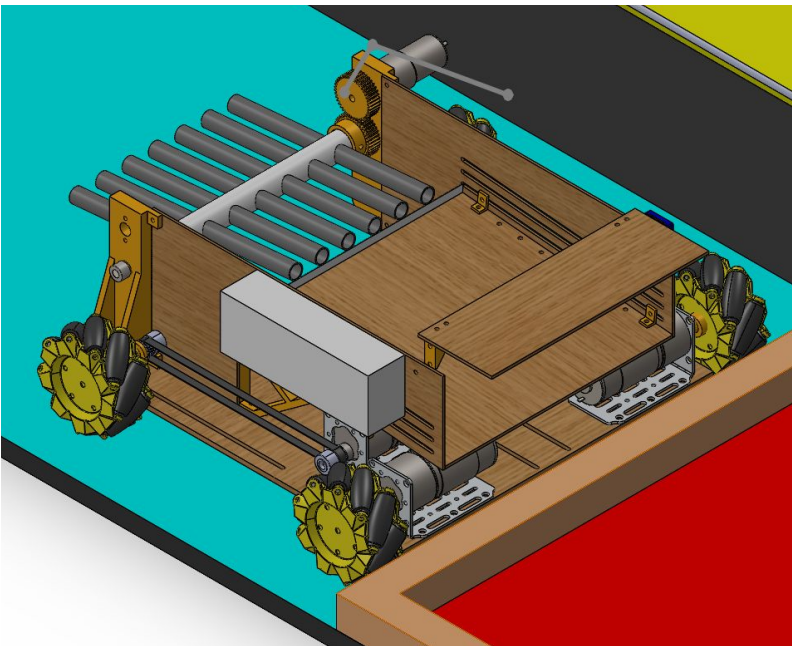


Figure 2.3.2.1
Original Design CAD

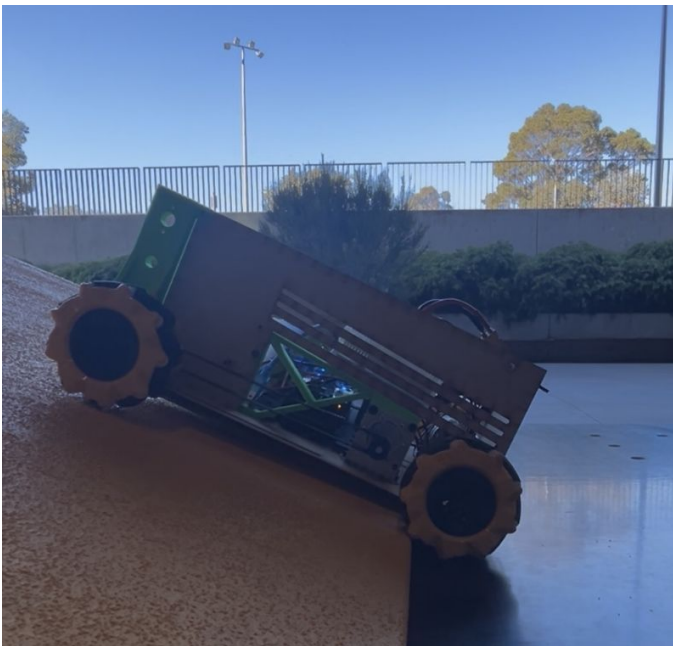


Figure 2.3.2.2
Seesaw Testing



Figure 2.3.2.4
Non-Slip Rug

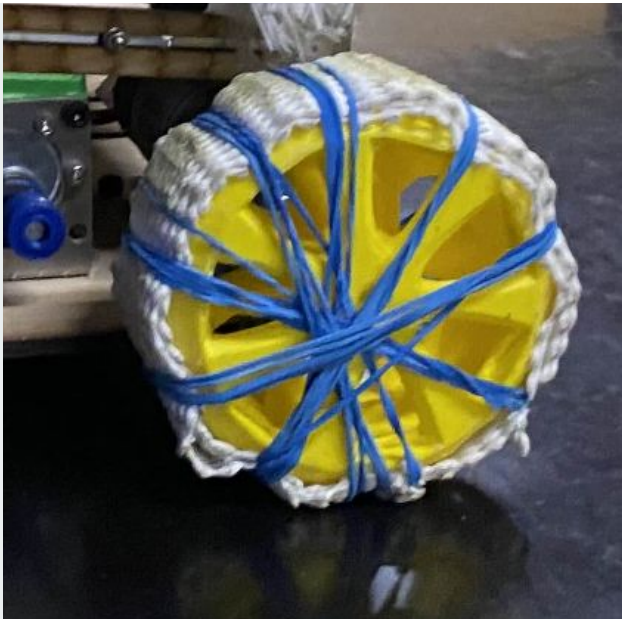
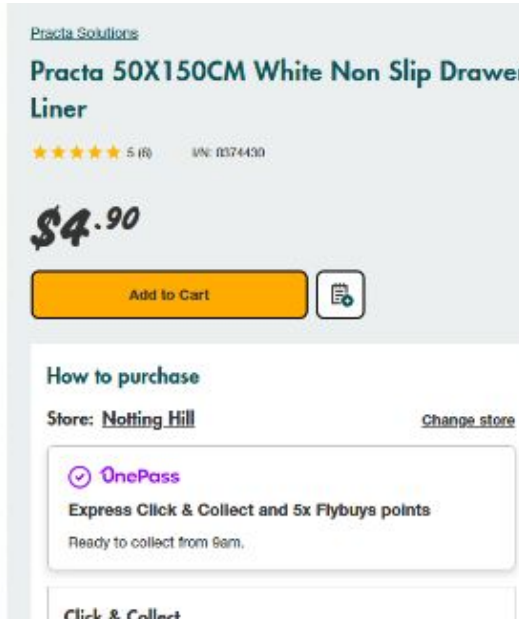


Figure 2.3.2.5
Final Design

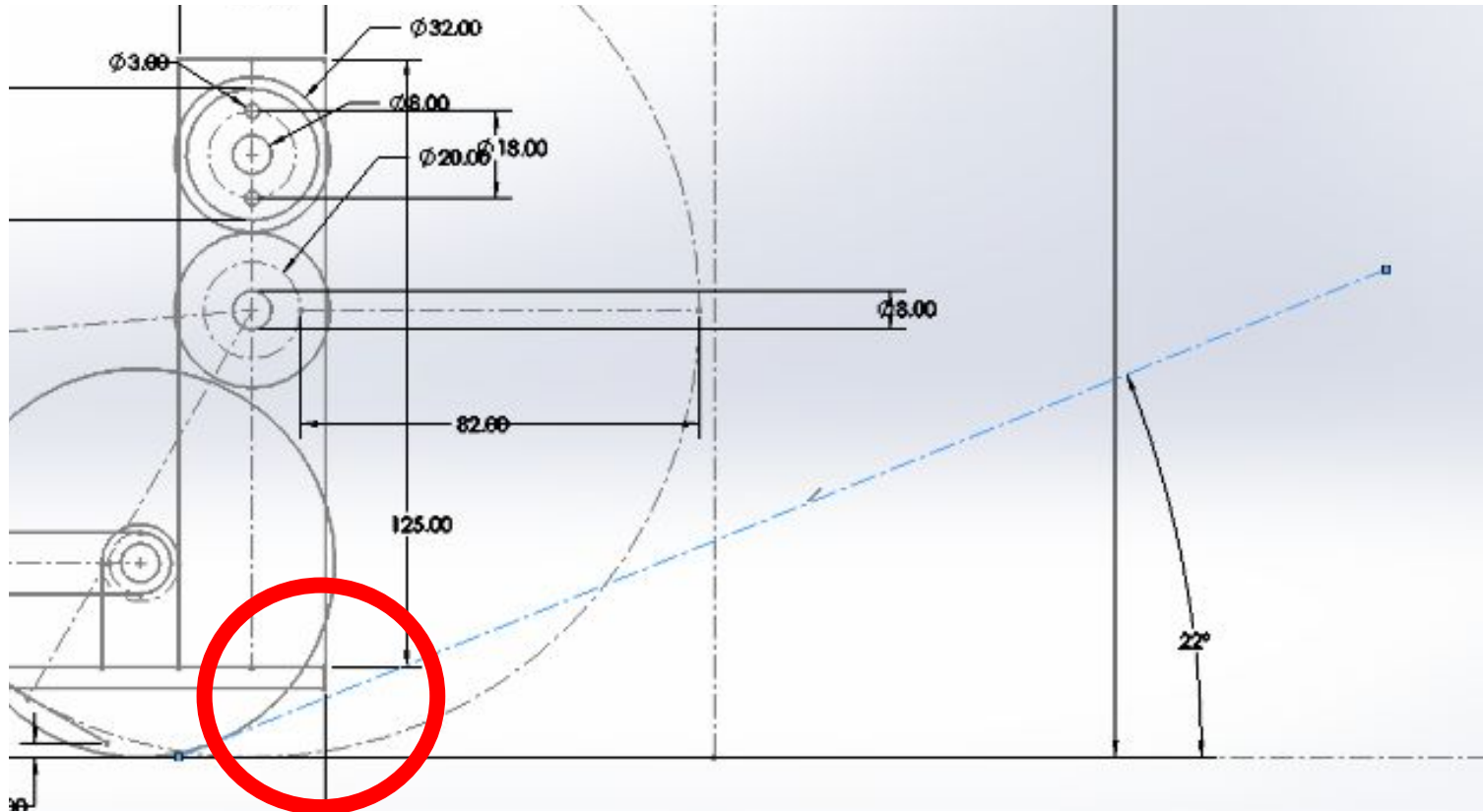


Figure 2.3.2.3
SolidWorks Draft - Wrong Assumption of the Seesaw

3. Project Reflection

Competition Performance

On the day of competition our best run managed to collect all three balls and deposit two which yielded a good score 135. However in our testing we recorded a full complete run which acted as a video submission demonstrating our robot depositing all 3 balls and completing the track in 36 seconds to score 222 points, placing us in 6th place overall

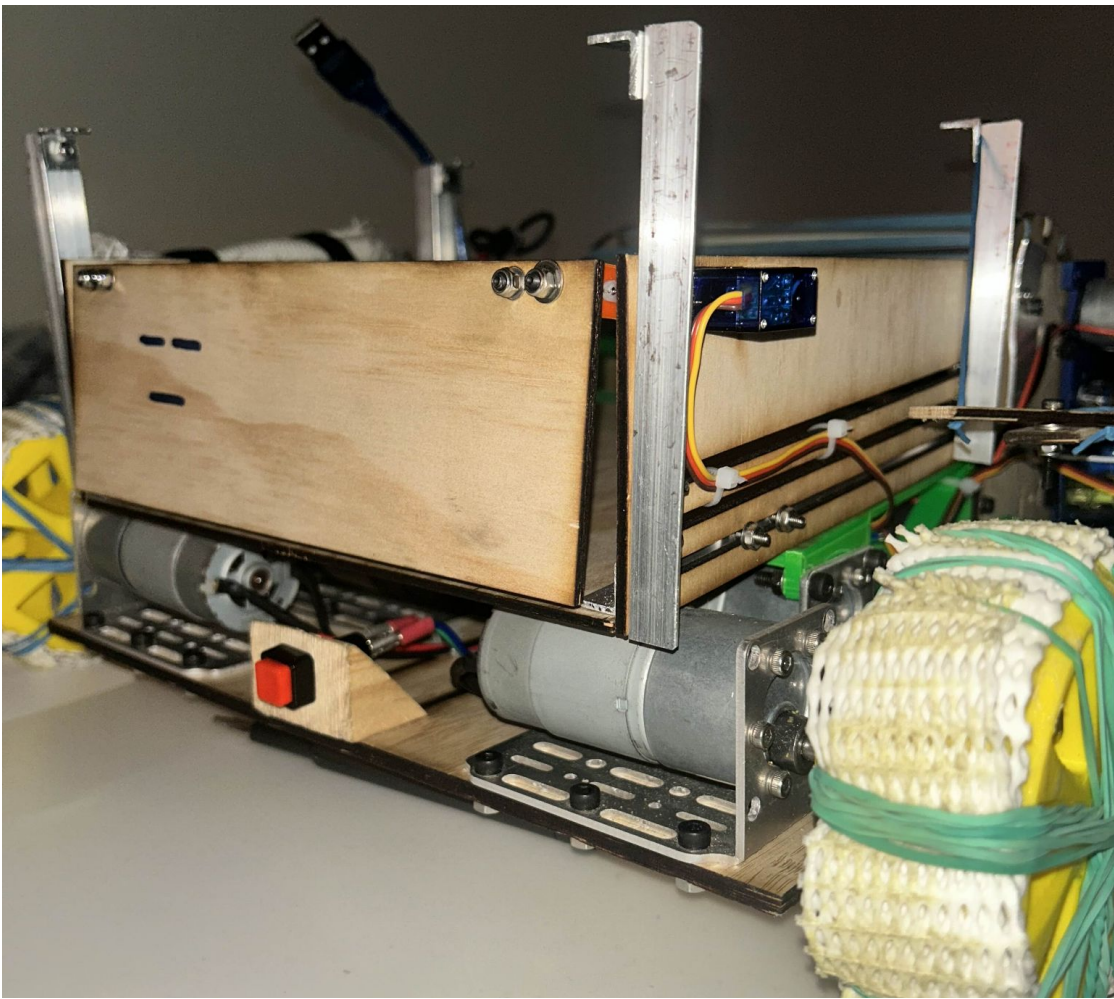


Figure 3.1
Roof Construction

Insights

Design / Analysis

Once put into the group and given the project it was unanimous that we wanted to get started early and quickly went off to produce a design each before voting on which solution to pursue. We voted based on the practicality and attainability of the designs and ultimately chose a spinning collection tool which provided us with a great start to the project.

Project Planning/timelining

We approached the project by planning what we would like to achieve within a week (before the next class) and setting that as the goal. This worked well initially with a loosely scaffolded plan to have a prototype for movement testing but we lacked a clear timeline to take us up to the competition. It wasn't until close to the competition day that we felt the ramifications of this approach, which became a scramble to adjust and fine tune the robot.

Testing and Development

Since we had a design chosen early we quickly got to realising the concept and started laser cutting and 3D printing to start producing the robot. We had a great head start and quickly built a platform to start testing with. In this testing we discovered unforeseen issues with the collection device. To solve this problem we tried many different designs and materials to find one that worked but this consumed weeks of our time. This shows that concepts and theory are a great start but practice is a true test that exposes flaws very quickly. This greatly slowed down the progress that we had made forced us to cram many hours in the last week and a half to get the robot operational as other areas of the robot needed to be worked on.

Team Management / Communication

We messaged on instagram and called on discord to ensure that all members knew what was expected of them and where we were tracking as a team it was a great resource especially to inform members who couldn't make it to meetings of what they missed. It was hard to manage the team effectively and would often lead to time spent idle as we would often all try to work on one aspect of the robot, naturally only one or two people can actually be working on the robot due to the scale and fiddly nature of it which resulted in inefficient usage of time.

Experience gained

Collaborating in a team can be incredible beneficial when done correctly and this project as given us insight into how to better act as a team in future. Setting out a clear, attainable and detailed timeline to ensure consistent progress which reduces the rush as the due date approaches. This timeline would also help with task delegation which is an area we lacked in. We would all come together to try and come up with solutions to problems which essentially halted the progress until the problem was resolved. In future task delegation would avoid this issue by having a sub group try to resolve an issue while the rest of the team is still pushing ahead on other aspects of the project to maximise the time. Overall completing this project as taught us invaluable lessons that will be taken all the way through our engineering careers.

(604 words)

4. Reconciled Project Expenses

Payment Date	Item	Website/Store	Paid By	Price per item	Amount	Payment Amount	Running Total	Total per person
19/03/2025	Ardiuno Kit	Monash	Scofield	50.00	1	50.00	50.00	10.00
26/03/2025	80mm Mecanum Kit	Friend	Jason	0.00	1	0.00	50.00	10.00
02/04/2025	Second Hand Motor	Monash	Jake	22.50	1	22.50	72.50	14.50
09/04/2025	Plywood	Bunnings	Jake	15.20	1	15.20	87.70	17.54
09/04/2025	Hose	Bunnings	Jake	2.63	2	5.26	92.96	18.59
09/04/2025	Second Hand Motor	Monash	Nikhunj	22.50	3	67.50	160.46	32.09
09/04/2025	Motor Mounting Bracket	Monash	Nikhunj	5.00	4	20.00	180.46	36.09
10/04/2025	6mm Smooth Rod - 500mm	Maker store	Scofield	8.93	1	8.93	189.39	37.88
10/04/2025	Brass Flanged Bushes	Maker store	Scofield	1.45	6	8.70	198.09	39.62
10/04/2025	GT2 Closed Loop Timing Belt - 400mm	Maker store	Scofield	7.35	2	14.70	212.79	42.56
10/04/2025	Lock Collar - 6mm	Maker store	Scofield	1.58	2	3.16	215.95	43.19
10/04/2025	Shippment	Maker store	Scofield	12.00	1	12.00	227.95	45.59
12/04/2025	Medium motor	Jaycar	Scofield	23.95	1	23.95	251.90	50.38
28/04/2025	Hose Clamp	Bunnings	Jake	1.12	6	6.72	258.62	51.72
07/05/2025	Medium motor	Jaycar	Jason	23.95	1	23.95	282.57	56.51
09/05/2025	Silicone Tape	Bunnings	Jake	12.33	2	24.66	307.23	61.45
09/05/2025	Plumbing Sealant	Bunnings	Jake	17.72	1	17.72	324.95	64.99
10/05/2025	Drawer Liner	Bunnings	Jake	3.92	1	3.92	328.87	65.77
10/05/2025	Mercury Switch	Jaycar	Jake	2.75	1	2.75	331.62	66.32
10/05/2025	Push Button	Jaycar	Jake	2.45	1	2.45	334.07	66.81
12/05/2025	SG90 servo	Jaycar	James	11.95	2	23.90	357.97	71.59
28/05/2025	Monash Refund	Monash Shop	Nikhunj	(87.50)	1	(87.50)	270.47	54.09
Team Member		Amount Paid	Amount Owing					
Scofield		121.44	(67.35)					
Jake		101.18	(47.09)					
Nikhunj		0.00	54.09					
James		23.90	30.19					
Jason		23.95	30.14					

5. Conclusion

The final prototype was the result of various different designs, with many iterations being tested. This trial and error process was a constant throughout the development of each part of our final design, as there often were challenges when going from concept to physical prototype. The ball collection mechanism, for example, went through many such iterations before we finally landed on a reliable system that consistently collected the balls. Each time we hit a roadblock and needed a new design, the team had brainstorming sessions that forced us to innovate, keeping in mind to utilise the resources at our disposal as well as the previously manufactured parts.

While the project was undoubtedly challenging, we all agree that it was an enjoyable and stimulating experience. Each time we faced an issue whether mechanical, electrical or software-related, everyone had a creative solution to try and fix it. This gave us different approaches for the same problem and made problem-solving a lot more of an enjoyable and collaborative process. Sometimes, the solution to major problems were as simple as drilling a few more holes and securing things down, yet there would be highly engineered and creative solutions suggested, which shows that there was a growing understanding of the new manufacturing processes amongst the team.

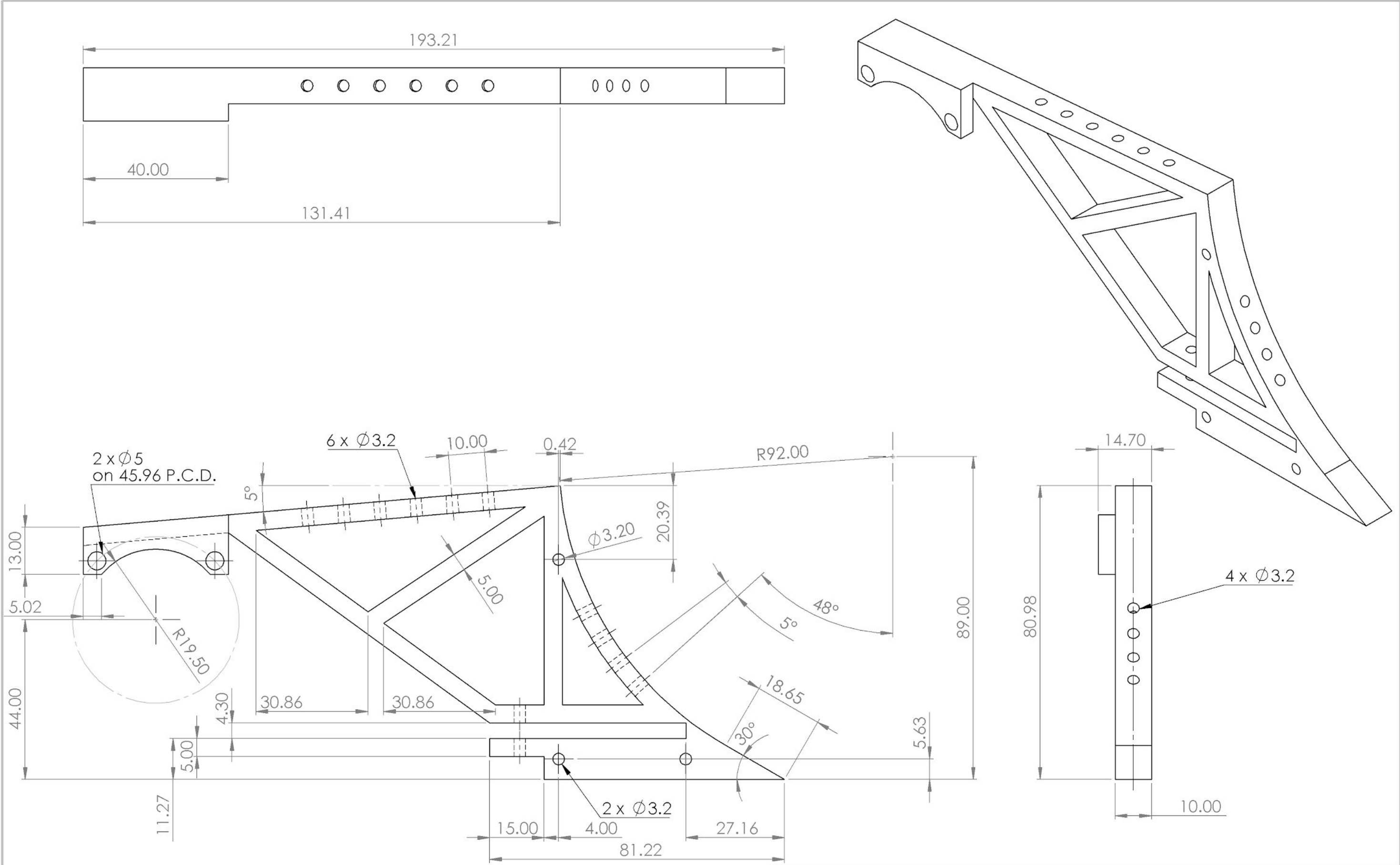
Throughout the process of designing and building the bot, we developed a range of valuable skills, from technical fabrication and CAD modeling to time management, teamwork and prototyping. These are all transferable skills that can prove beneficial in the future for both academic and professional projects.

Overall, this project has highlighted to us the importance of being able to adapt and make changes quickly to designs and systems. Like mentioned earlier, there were times when progress had stalled as mechanisms had failed or there were errors in our initial designs. In the event of such setbacks, we learned to quickly change designs or mechanisms and make informed decisions to rapidly change parts. These experiences strengthened our technical capabilities and ability to communicate.

Above all, we're proud of what we managed to achieve as a team, as at the very minimum we managed to build a functioning robot that completed the task successfully. Granted we have had discussions after the competition about different aspects we would have done differently in hindsight, but these these brainstorming sessions are what fueled the creation of the initial designs as well. Through this we have also built strong working relationships, and have likely formed friendships that will last well beyond this unit.

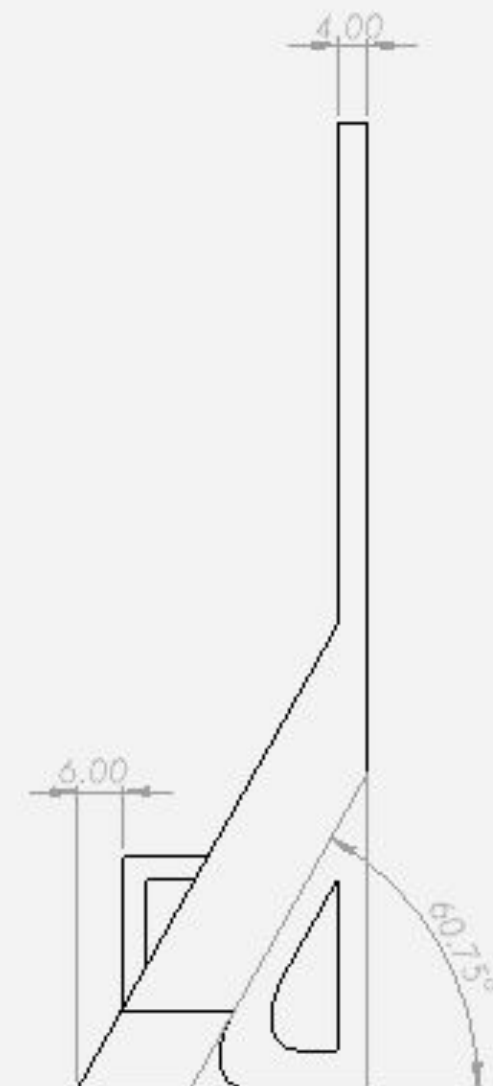
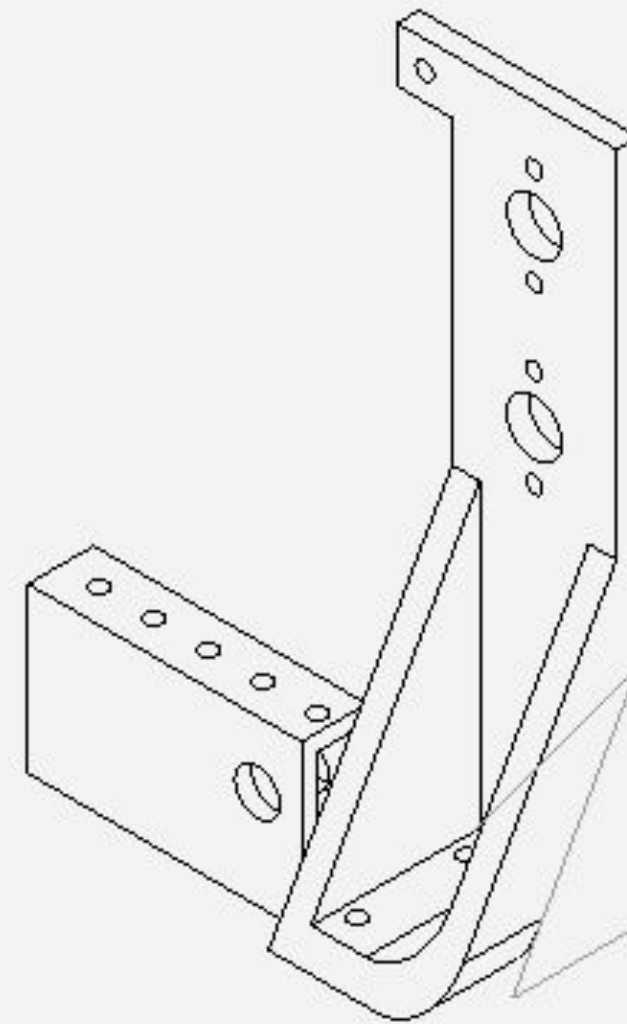
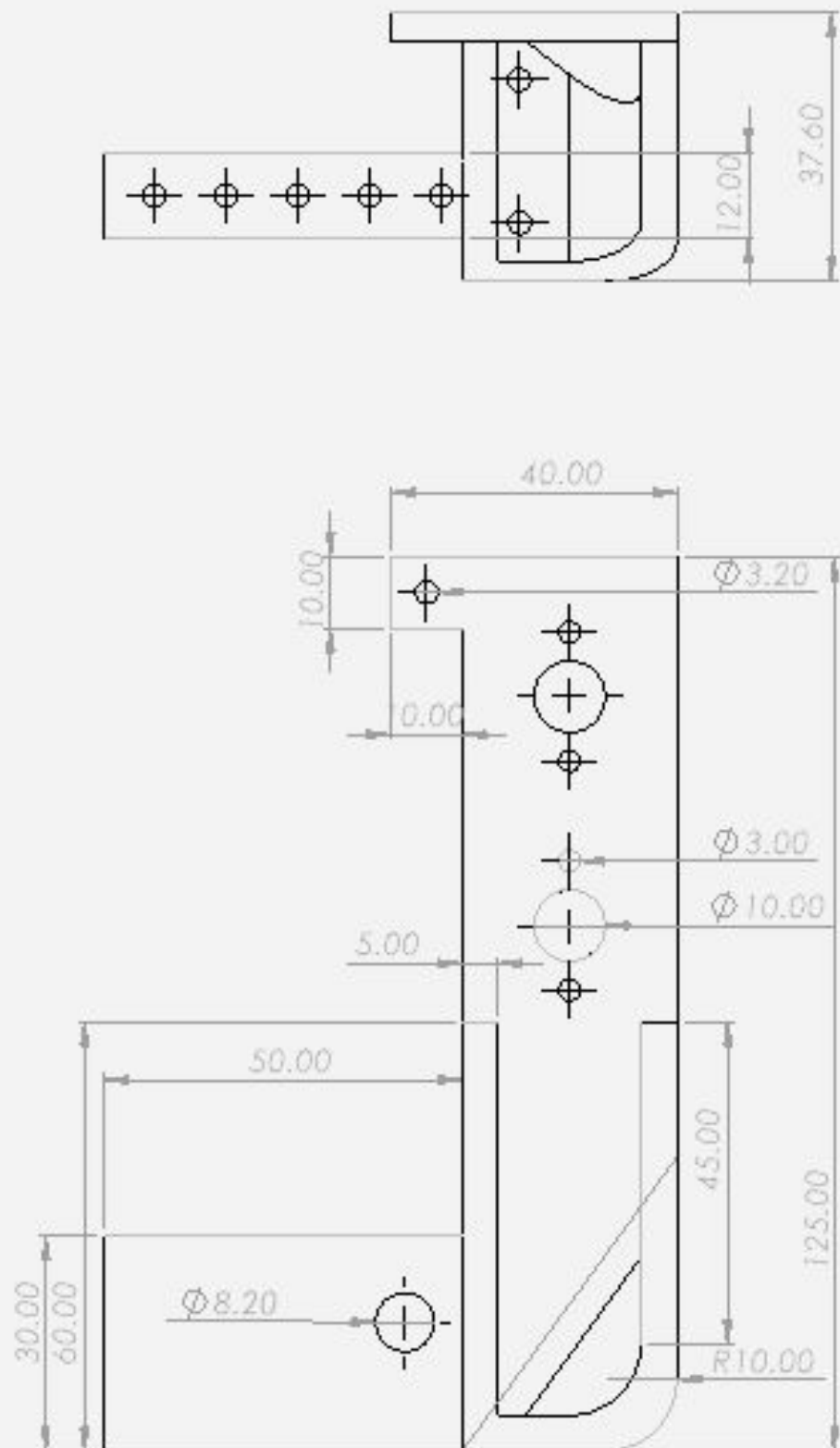
6. Appendix A: Engineering Drawing

Detail Drawing 1:



NOTES: DIMENSION IN mm GENERAL TOL: +0.2		TITLE: TEAM 77 - ENGINEERING DRAWING - STRUCTURE_V2	
	MATERIALS: PLA	MONASH UNIVERSITY DEPARTMENT OF MECHANICAL ENGINEERING	DATE: 22/05/2025
		DRN BY: Scofield Lai	A3 1:1 DWG NO. 33759146

Detail Drawing 2:



NOTES:

50

TEAM 77 - ENGINEERING DRAWING
INTAKE SYSTEM HOLDER V2

MATERIALS:
PLA

MONASH UNIVERSITY DEPARTMENT
OF MECHANICAL ENGINEERING

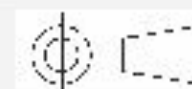
DRN BY: Jason Zhu

DATE: 20/02/2023

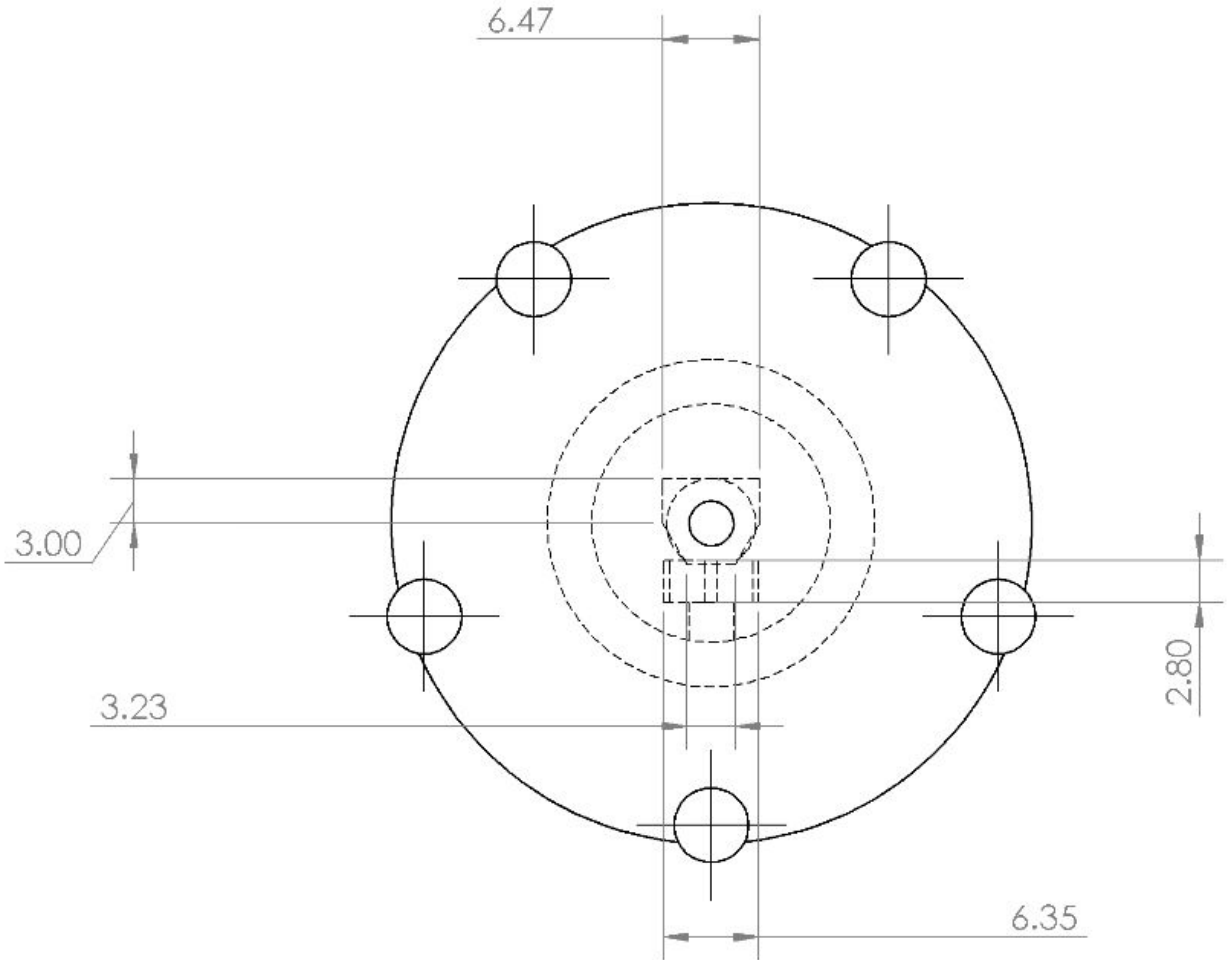
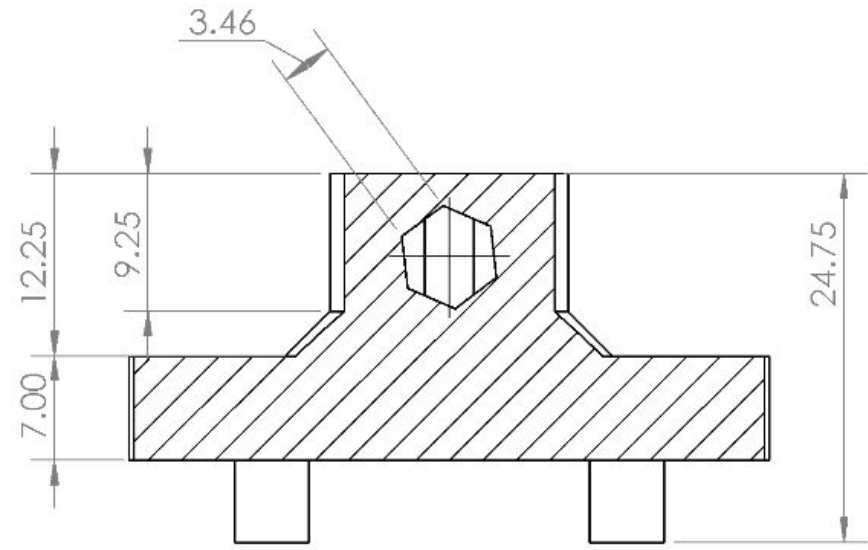
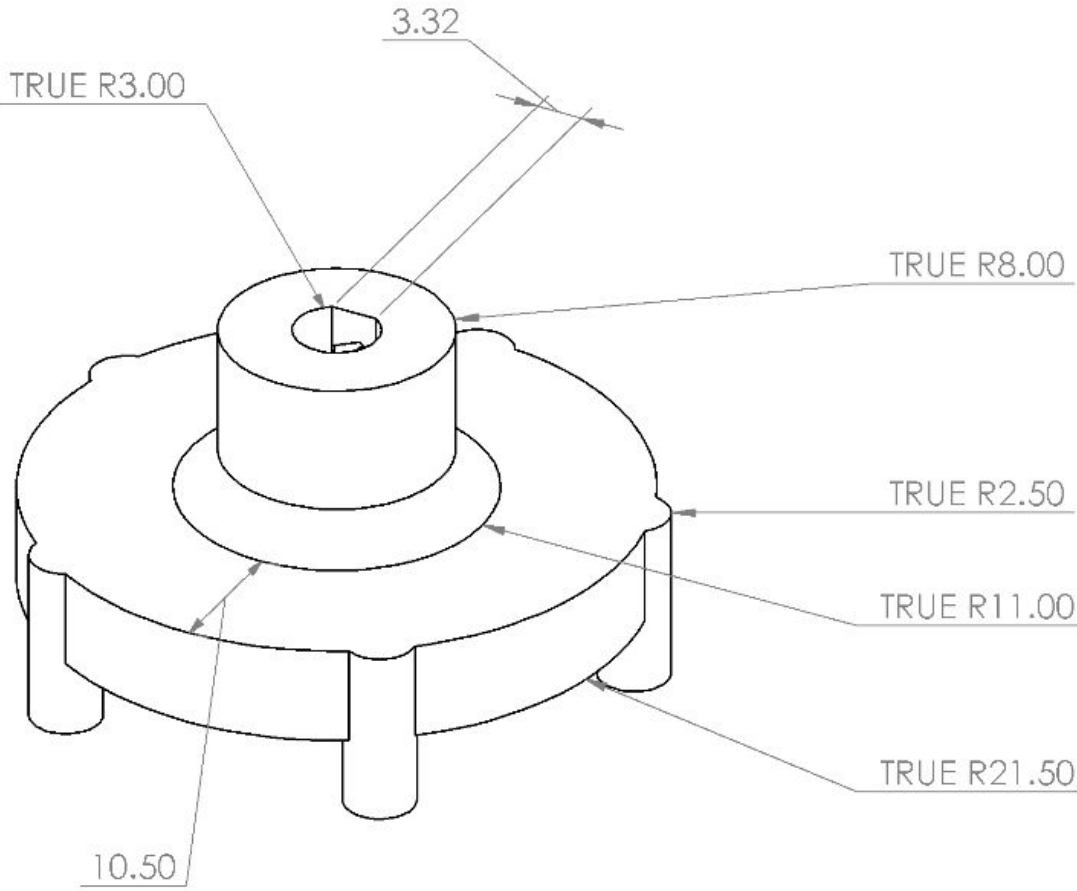
A3

SCALE: 1:1

DWG NO. 33855005

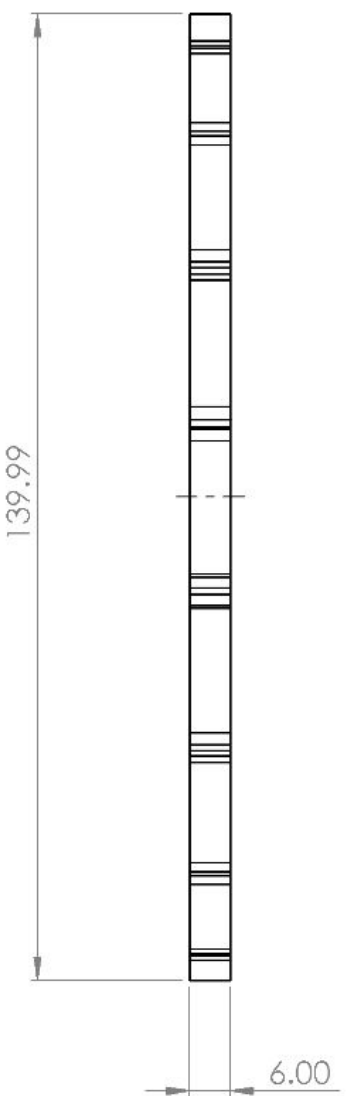
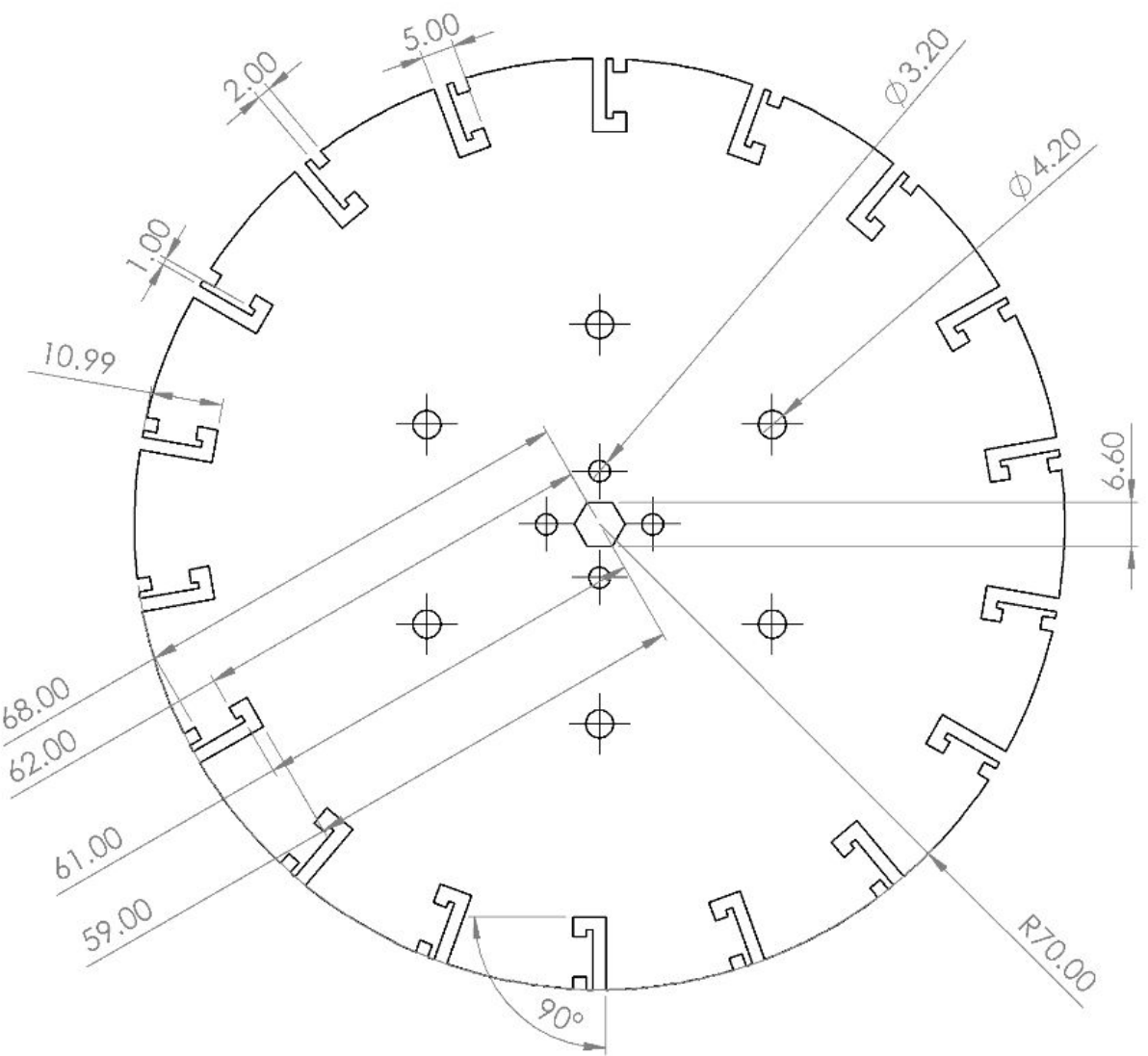


Detail Drawing 3:



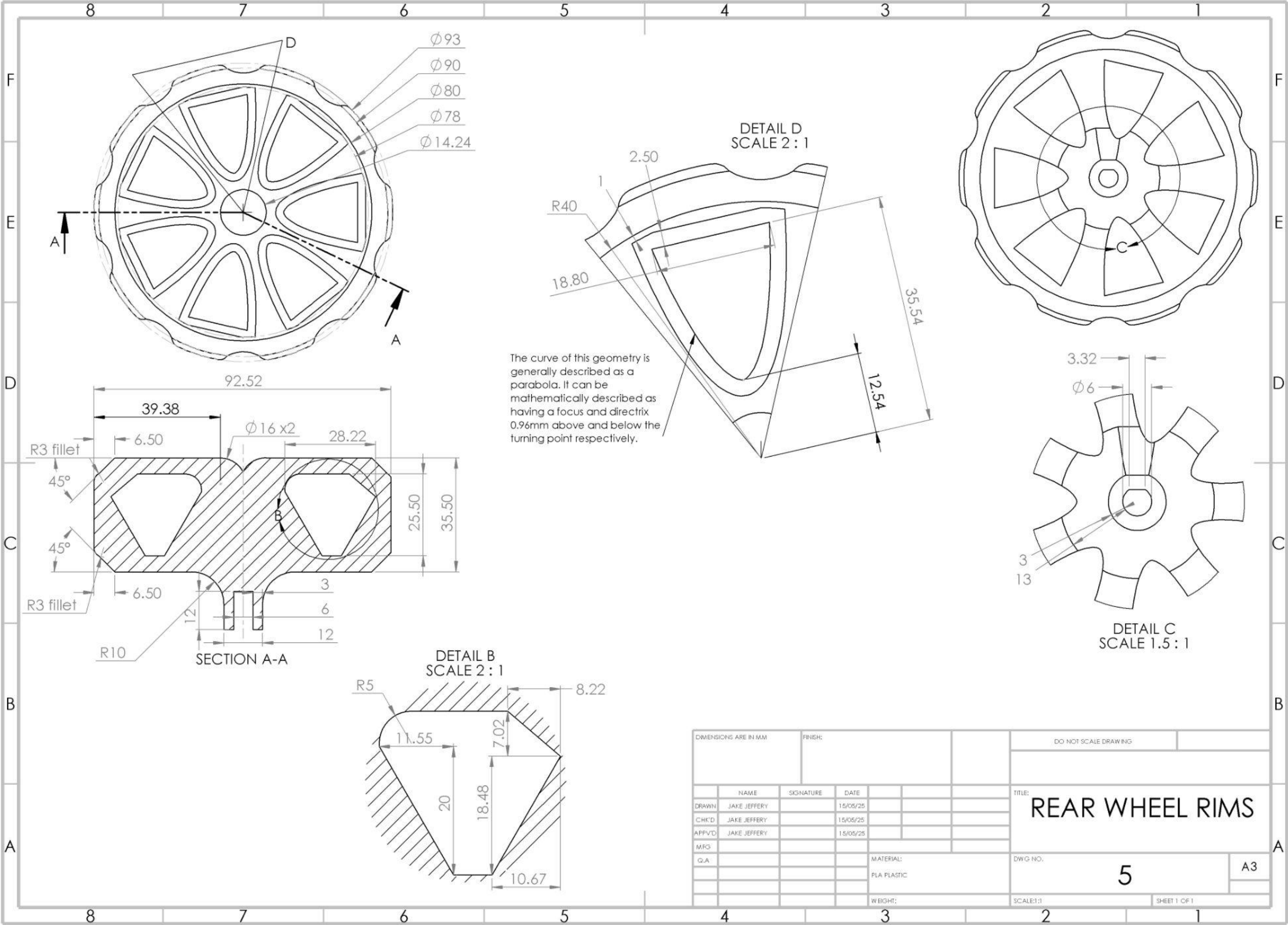
NOTES: DIMENSIONS IN mm GENERAL TOL: +/- 0.2		TITLE: TEAM 77 - ENGINEERING DRAWING - WHEEL CONNECTOR_V3		
	MATERIALS: PLA	MONASH UNIVERSITY DEPARTMENT OF MECHANICAL ENGINEERING	DATE: 25/05/2025	A3 SCALE: 2:1
			DRN BY: Nikhunj Ajay Nambiar DWG NO. 33369011	

Detail Drawing 4:



NOTES: Dimension in mm General tol: +0.2	TITLE: Team 77-Roller		Roller_T1	
	MATERIALS: Plywood	MONASH UNIVERSITY DEPARTMENT OF MECHANICAL ENGINEERING	DATE: 26/11/25	A3
			DRN BY: James Robinson	SCALE: DWG NO. 34988041

Detail Drawing 5:



7. Appendix B: Meeting Minutes

MEC2402/MAE2412	WEEKLY STAND-UP MEETING	DATE	Week 4	START	2:34	TEAM #	77
Team Member (Mark any APOLOGIES, LATE or ABSENT)	What you were you SPECIFICALLY tasked to do last week? If the team allocated no formal tasks write NOTHING.	WHAT did you achieve last week? If it is not on the team drive, it is not done, write NOTHING.	WHAT does the team need you SPECIFICALLY to do in the next week? (must not be the same answers for others), and WHEN is it DUE	WHO is responsible for checking your work is completed and meets requirements (must be an individual)			
Jason	Isometric drawing	Isometric drawing	OFFERS, CSG. DM				
Jake	Isometric drawing	Isometric drawing	Arduino Tutorial				
James	Isometric drawing	Isometric drawing	OFFERS, CSG. DM				
Nick	Isometric drawing	Isometric drawing	Arduino Tutorial				
Scofield	Isometric drawing	Isometric drawing	CAD				

7. Appendix B: Meeting Minutes

MEC2402 / MAE2412		TEAM MEETING				DATE	02/04/2025	START TIME	2:17
TEAM #	77	Team Work Reflection				VENUE	Computer Lab	FINISH TIME	2:34
1: ATTENDANCE									
ON-TIME									
James									
Nikhunj									
Jake									
Jason									
Scofield									
APOLOGIES									
LATE									
NO SHOWS									
2: MEETING ROLES									
TIMEKEEPER									
Jake									
SCRIBE									
James									
FACILITATOR									
Jason									

7. Appendix B: Meeting Minutes

MEC2402		Meeting Date	09/04	Meeting Time	2:30
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Team members:	Jake, James, Nikh, Jason, Scofield
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Task for this week					
Copy and Paste from last week's minutes				Team to Update BEFORE Studio	
# or Set	Description of Task	Measurable Deliverable(s)	Team Members (Lead / Supporting)	Notes on review of Deliverables	Delivered? (%)
	Manufacturing (Laser cut)	Cut all board needs	ason, Scofield, Jame		100%
	Manufacturing (3d printing)	3D print of component	Scofield		70%
	Assembly	Assembly of the robot	ason, Scofield, Jame		50%
	Manufacturing (Drill)	Drill for water pipes	ason, Scofield, Jame		50%
	Ardiuno and prototype		Nikh and jake		50%

Week	Week 6
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Team	77
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Plan for the Next Week (ONLY)			
Team to Meet and Update In the Studio			
# or Set	Description of Task	Measurable / Tangible Deliverable(s) (what evidence is required to ensure the task has been completed?)	Team Members (Lead / Supporting)
	Order belt, rod		scofield
	prototype assembly		jake and Nikh

7. Appendix B: Meeting Minutes

MEC2402		Meeting Date	30/4/2025	Meeting Time	2:23
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Team members:	Jake, James, Nikh, Jason, Scofield
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Task for this week					
Copy and Paste from last week's minutes				Team to Update BEFORE Studio	
# or Set	Description of Task	Measurable Deliverable(s)	Team Members (Lead / Supporting)	Notes on review of Deliverables	Delivered? (%)
	Working on ball intake system	Reliability in picking up balls	Scofield, Nikhunj, Jake	Make sure to get something working. Cannot afford to waste more time on this	75%
	Working on ball wiper	Reliability in pulling balls into direction of travel	Jason, James	Make sure that it can fit within the starting zone	50%
					50%
					50%
					50%

Week	Week 8
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Team	77
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Plan for the Next Week (ONLY)			
Team to Meet and Update In the Studio			
# or Set	Description of Task	Measurable / Tangible Deliverable(s) (what evidence is required to ensure the task has been completed?)	Team Members (Lead / Supporting)
	Get started on programming and testing on track	Get the first two phases completed	All

7. Appendix B: Meeting Minutes

MEC2402		Meeting Date	07/04/2025	Meeting Time	2:05
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Team members:	Jake, James, Nikh, Jason, Scofield
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Task for this week					
Copy and Paste from last week's minutes				Team to Update BEFORE Studio	
# or Set	Description of Task	Measurable Deliverable(s)	Team Members (Lead / Supporting)	Notes on review of Deliverables	Delivered? (%)
	Programming and testing the robot	How many phases can be completed with all three balls	All	Video each attempt as we are getting closer	90%
					50%
					50%
					50%
					50%

Week	Week 9
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Team	77
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Plan for the Next Week (ONLY)			
Team to Meet and Update In the Studio			
# or Set	Description of Task	Measurable / Tangible Deliverable(s) (what evidence is required to ensure the task has been completed?)	Team Members (Lead / Supporting)
	Prepare for the competition	Achieve a full mission complete	All